



ADDIS ABABA UNIVERSITY

COLLEGE OF NATURAL AND COMPUTATIONAL

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DEPARTMENT OF COMPUTER SCIENCE

CHARITY MANAGEMENT SYSTEM

FINAL YEAR PROJECT I

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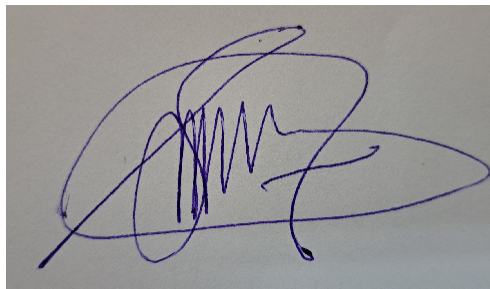
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ACRONYMS

FR : Functional Requirement

NFR : Non-Functional Requirement

PR : Pseudo Requirement

UI : User Interface

API : Application Programming Interface

CSS : Cascading Style Sheets

ODM : Object Data Modeling

ID : Identifier

UC : Use Case

SC : Scenario

HTTPS : Hypertext Transfer Protocol Secure

SSL : Secure Sockets Layer

WCAG : Web Content Accessibility Guidelines

DB : Database

NoSQL : Not Only Structured Query Language

JSON : JavaScript Object Notation

IDE : Integrated Development Environment

URL : Uniform Resource Locator

API : Application Programming Interface

RBAC : Role-Based Access Control

UI/UX : User Interface / User Experience

IT : Information Technology

CHAPTER ONE

PROJECT PROPOSAL

1.1 Introduction

Charity organizations play an important role in supporting vulnerable communities, yet many of them in Ethiopia still depend on outdated and mostly manual methods to manage donations, record financial transactions, and communicate with donors or beneficiaries. These practices often involve paper forms, scattered spreadsheets, and unstructured updates through phone calls or social media. As a result, important information may be lost, misrecorded, or delayed, and there is no consistent way to monitor how much a campaign has collected or how the collected resources were distributed.

A charity management system introduces a structured and organized way to handle these operations. By providing a web-based platform, charities can present their campaigns more clearly, donors can give with confidence, and administrators can oversee the entire process with proper records.

The system supports features such as campaign creation, donation tracking, donor anonymity, and beneficiary recording. It serves as a digital environment where all participants, donors, charity staff, and administrators interact through consistent and verifiable information. This project focuses on designing and developing such a system to minimize inefficiencies, improve transparency, and strengthen the overall trust surrounding charitable activities.

In general, this project aims to simplify and organize how charitable organizations manage their activities. By providing a single platform for donation tracking and campaign management it reduces errors, saves time, and ensures information is clear and reliable. The system helps charities operate more efficiently and transparently, allowing them to focus on supporting those in need.

1.2 Statement of the Problem and Justification

In many local charity settings, the absence of a unified management system creates significant operational challenges. Donation records are often maintained on paper or stored informally, making them prone to loss, damage, or inconsistencies. Donors rarely receive updates on how their contributions are utilized, and campaign organizers spend

considerable time manually calculating collected amounts, tracking remaining needs, and preparing reports for supervisors or supporters. The scattered and unverifiable nature of this information can lead to donor hesitation, as concerns about misuse or unclear fund distribution persist.

Traditional manual processes in these charity organizations are therefore inefficient, time-consuming, and prone to errors. This not only affects the trust and engagement of donors but also limits the ability of charities to manage campaigns effectively and transparently.

Introducing a centralized online charity management system addresses these challenges by providing a stable, auditable, and accessible digital platform. Such a system can record every donation, specify its purpose, and link it to a particular campaign, reducing errors and misunderstandings. Donors gain confidence through transparent progress updates, while charities benefit from decreased administrative burden and improved efficiency. Features such as anonymous donations further enhance inclusivity and participation.

In summary, the problem lies in the fragmented and unreliable methods currently used to manage charitable activities, underscoring the need for a web-based system that ensures transparency, accountability, and efficient management, ultimately fostering greater donor trust and sustainable charitable engagement.

1.3 Project Objective

1.3.1 General Objective

The general objective of this project is to design and develop a web-based charity management system.

1.3.2 Specific Objectives

To achieve the general objective, the project will focus on the following specific objectives:

- Conduct data collection to understand the requirements of the system.
- Analyze the collected data to identify key features and functionalities.
- Design the proposed system architecture.

- Identify appropriate tools and technologies for the design and implementation of the system.
- Implement the system according to the design specifications.
- Test and evaluate the system to ensure it functions correctly and meets project requirements.

1.4 Scope and Limitation of the Project

1.4.1 Scope of the project

- The system provides an online platform that facilitates interaction between donors, charity organizations, and administrators through a web interface.
- It enables charity organizations to register, manage campaigns, record donations, update progress, and track beneficiaries.
- Donors are able to browse campaigns, make contributions, choose anonymity, and receive digital confirmations.
- Administrators are given tools to verify charity organizations, monitor activities, and generate basic reports.
- All information is stored in a centralized database designed to maintain consistency and improve traceability.
- The platform focuses on offering a clear and structured flow of activities that replaces traditional paper-based or informal record-keeping methods.

1.4.2 Limitations of the Project

- The system does not include live integration with financial payment gateways; donations are recorded conceptually for demonstration purposes.
- It does not implement automated verification methods such as ID checking or fraud detection, relying instead on manual oversight.
- The project is limited to a browser-based environment and does not include dedicated mobile applications.

- It does not attempt to cover legal or tax-related procedures associated with charity operations.
- Advanced features such as multi-currency handling, or predictive analytics fall outside the intended scope of this version.

1.5 System Development Methodology

The Agile Software Development Methodology is used for this project because it supports iterative development, flexibility, and continuous feedback. It allows requirements to be refined gradually and supports documentation and implementation.

1.5.1. Investigation (Fact-Finding) Methods

To gather accurate and relevant information for the development of the charity management system, multiple fact-finding techniques were used:

- **Interview** : Discussions were held with people familiar with charity activities, including charity organizers, donation coordinators, and potential donors. These interviews helped identify challenges such as lack of transparency, difficulty tracking funds, and poor communication between donors and beneficiaries.
- **Document Review** : Existing donation forms, financial logs, and social media postings were reviewed. This helped identify gaps in manual record-keeping.
- **Survey**: A survey was conducted with potential donors and charity workers to gather insights on donation preferences, transparency, anonymity, and campaign tracking. The results helped identify challenges and informed the design of the charity management system.

1.5.2. System Development Tools

1.5.2.1 Front-End Development Tools

Tool	Version	Purpose
------	---------	---------

React	19.2.4	Used to build the application's user interface, manage state, and enable dynamic user interactions using a component-based architecture.
Tailwind CSS	4.1.18	Used for styling the user interface with utility-first CSS, enabling responsive design and consistent styling without writing custom CSS.

Table 1.1 Front-End Development Tools

1.5.2.2 Back-End Development Tools

Tool Name	Version	Purpose
Node.js	25.5.0	Serves as the JavaScript runtime environment for executing server-side code and handling scalable, event-driven operations.
Express.js	5.2.1	Provides a lightweight web framework for handling routing, middleware, server logic, and API endpoint management.

Table 1.2 Back-End Development Tools

1.5.2.3 Database Development Tools

Tool Name	Version	Purpose
-----------	---------	---------

MongoDB	8.2.3	Used as a NoSQL document-oriented database to store and manage data such as donors, campaigns, donations, beneficiaries, and transaction history.
Mongoose(ODM)	9.1.5	Used as an Object Data Modeling (ODM) library to define schemas, validate data, and manage interactions between the Node.js application and MongoDB.

Table 1.3 Database Development Tools

1.5.2.4 Other Development Tools

Category	Tool Name	Version	Purpose
Design & Modeling	Figma	Latest(Web-based)	Design UI, wireframes, mockups, and user flow prototypes
Development Environment	Visual Studio Code	Latest	Source code editing and project management
Development Environment	Local Development Environment	Node.js 25.5.0, MongoDB 8.2.3	Local testing and development before deployment
Testing Tool	Postman	Web-based	Testing backend API endpoints
Documentation Tool	Google Docs	Web-based	Writing, editing, and collaborative documentation

Version Control	GitHub	Web-based	Source code management and version control
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Table 1.4 Other Development Tools

1.6 Significance of the Project

Societal Benefits:

- Promotes transparency by allowing donors to track how their contributions are utilized, building trust between donors and charity organizations.
- Improves accountability, enabling charities to maintain accurate records and report progress efficiently.
- Ensures beneficiaries receive organized and verified support, helping aid reach those who need it most.
- Encourages donations by providing easy online access to donation platforms.
- Enhances privacy through anonymous donation options, allowing donors to contribute comfortably.

Technological Benefits:

- Facilitates digital transformation of charity operations, replacing manual record-keeping with an automated online system.
- Improves data management by reducing errors and duplication through centralized storage.
- Enhances security by safeguarding sensitive information via authentication and secure transaction mechanisms.
- Provides scalability to accommodate a growing number of users, campaigns, and functionalities.
- Offers real-time tracking for donors and administrators, ensuring timely monitoring of campaign progress.

- Makes charitable activities more efficient, transparent, and impactful through technology.

1.7 Beneficiaries

Charity Organizations

- Efficiently manage donation campaigns.
- Track contributions and generate detailed reports.
- Communicate effectively with donors and beneficiaries.

Donors

- View available campaigns and make secure donations.
- Choose to donate anonymously.
- Monitor past contributions and track the impact of their support.

Government

- Supervision of charity organization by the government body will be easier since it can find multiple charity organization data in one place.
- Can help in reviewing campaign activities in order to prove the legitimacy by seeing the revenue that campaign generated in a transparent way.

1.8 Time Schedule

Gantt Chart

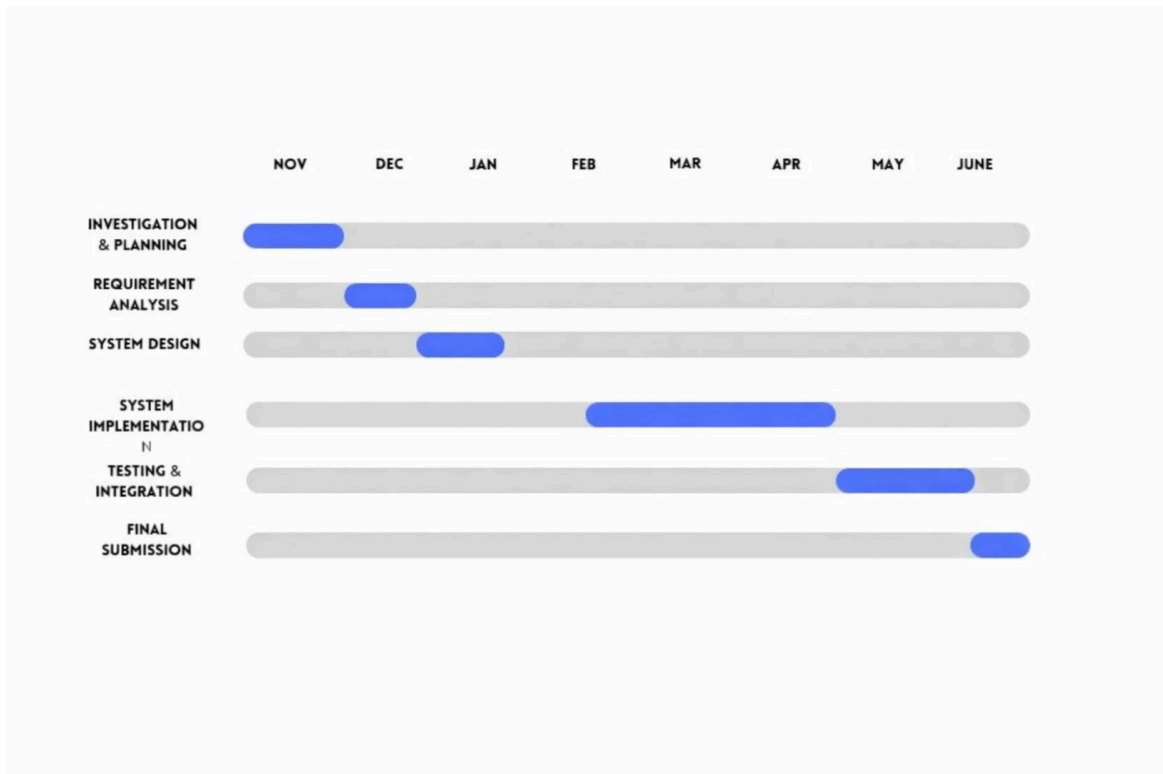


Figure 1.1 Gantt Chart

CHAPTER TWO

REQUIREMENT ANALYSIS

2.1 Introduction

The requirement analysis section forms the conceptual foundation of the proposed charity management system by establishing a clear and structured understanding of what the system must accomplish, how it is expected to function, and the constraints under which it will operate.

This chapter plays a crucial role in the system development process as it ensures that the planned solution does not emerge solely from technical assumptions but directly responds to real-world challenges faced by charity organizations, donors, and administrative bodies. By systematically examining the shortcomings of the current charity management landscape, this analysis bridges the gap between user expectations and practical system design, thereby offering a coherent roadmap for implementation.

In many communities, charitable activities play an essential role in supporting vulnerable populations, funding social projects, and sustaining community welfare initiatives. However, the operational environment within which these activities take place is often fragmented, heavily manual, and lacking the transparency necessary to inspire donor confidence.

Campaign announcements are dispersed across informal channels, donation tracking lacks consistency, and reporting mechanisms rely on handwritten notes or improvised documentation. These inefficiencies not only slow down the workflow but also create opportunities for miscommunication, mistrust, and data loss.

The proposed charity management system aims to address these challenges by digitizing the core functions involved in organizing and overseeing charitable campaigns. Designed as a web-based platform, the system will provide a centralized environment where donors can easily access verified campaigns, make secure contributions, and optionally remain anonymous. Charity organizations will be equipped with structured tools for creating campaigns, logging donations, presenting progress updates, and preparing financial reports. Administrators will oversee platform activities through an integrated control

panel that enhances accountability and ensures the legitimacy of participating organizations.

This requirement analysis identifies both the functional components of what the system must do and the non-functional elements how well the system must perform. It ensures that each proposed feature aligns with the broader objectives of transparency, efficiency, and trust-building within the charitable ecosystem.

2.2 Current System

Despite the increasing societal need for transparent, efficient, and accountable donation management, the operational environment remains largely informal and fragmented. Most charitable initiatives operate using manual, paper-based processes or ad hoc digital tools that lack centralization, accuracy, and long-term reporting capabilities.

The analysis below outlines the key operational workflows in the current environment, identifies the main stakeholders engaged in charity work, and highlights the tools and communication channels commonly relied upon. By examining the existing system in depth, this section clarifies the inefficiencies and risks that necessitate a structured digital platform.

2.2.1 Major Functions of the Current System / Current System Description

1. Campaign Announcement and Promotion

Most charities communicate new campaigns using informal channels such as printed posters, social media posts, word of mouth, or messaging apps. There is no centralized digital dashboard where donors can browse ongoing activities, verify legitimacy, or track previous events. As a result, campaign visibility depends heavily on human networks and inconsistent outreach.

2. Donation Collection and Recording

Donations are typically collected in person through cash handover, bank transfers, or mobile money payments shared directly with organizers. These transactions are documented manually through: handwritten notes, simple receipt books, personal notebooks of volunteers, scattered Excel sheets saved individually.

Because there is no consolidated database, donor records are often incomplete, duplicated, or misplaced.

3. Report Preparation and Accountability

Financial reports and summary documents are prepared manually by compiling receipts, tallying contributions, and summarizing activities. This process is slow, error-prone, and highly dependent on individual volunteers. Stakeholders, including donors, rarely receive consistent or verifiable updates.

Typical Workflow in the Current System

1. A campaign is announced informally through posters, messages, or social media.
2. Donors contact organizers and send cash or transfers manually.
3. Organizers record donations on paper or spreadsheets.
- 4 Funds or goods are delivered in person.
- 5 Organizers compile handwritten notes to produce a basic report.

This workflow lacks automation, security, and long-term storage.

2.2.2 Problems of the Existing System

1. Lack of Transparency and Trust

Manual record-keeping creates opportunities for errors, loss of information, and even suspicion of mismanagement. Donors have no way to verify whether funds were used as intended.

2. Fragmented and Inconsistent Data

Donation records are scattered across paper notebooks, screenshots, and spreadsheets. This fragmentation prevents efficient tracking of donor history or campaign performance.

3. Time-Consuming Reporting

Creating financial reports requires manually checking receipts and summarizing totals. This leads to delays and increases the risk of mistakes.

4. High Risk of Data Loss

Paper receipts can be damaged or lost; personal phones containing screenshots can break or be formatted. There is no reliable backup system.

5. No Centralized Communication

Important updates are shared across multiple channels, making coordination difficult and sometimes leading to misunderstandings.

6. Limited Donor Engagement

Donors contribute once but rarely receive structured updates, progress trackers, or impact reports that would encourage repeat donations.

7. Inefficient Workflow for Organizers

Volunteers spend significant time: rewriting donation lists, sorting receipts, manually contacting donors, updating spreadsheets.

This reduces efficiency and increases stress during high-volume campaigns.

2.3 Requirement Gathering

The development of the charity management system requires a structured and thorough understanding of the operational needs, existing challenges, and expectations of the individuals and groups involved in managing donation-related activities. The requirement-gathering process examined current charity workflows, observed how information is recorded and communicated, and analyzed practices commonly used by local charity initiatives. Since most existing processes rely on manual, non-standardized tools, this phase focused on identifying functional gaps and determining essential features needed to build an organized, transparent, and user-friendly digital platform.

2.3.1 Requirement Gathering Methodologies

Requirement-gathering methodologies for the charity management system utilized a combination of research and stakeholder oriented techniques to ensure a clear and comprehensive understanding of user needs.

Online Research Across Social Media and the Web

Research was conducted on social media platforms, charity websites, and online fundraising pages to understand how local organizations present campaigns, manage

donations, and interact with donors in a digital environment. This helped reveal common practices, user expectations, and challenges such as campaign visibility, donor feedback, and confirmation mechanisms. Insights gathered highlighted essential features such as donation tracking, campaign transparency, and customizable donor anonymity.

Review of Existing Documents and Records

Existing physical and digital materials including receipt books, handwritten donor lists, Excel sheets, and social media announcements were analyzed to understand what information is currently collected and how it is maintained. This review exposed inconsistencies in data formats, missing donor details, irregular reporting, and the absence of standardized templates for tracking campaign goals, collected amounts, and remaining balances.

Analysis of Common System Patterns

A brief analysis of common digital charity and fundraising platforms was performed to identify standard system patterns that could inform the design of the proposed solution. Features such as structured campaign listings, donor dashboards, secure donation logging modules, and administrative monitoring tools were particularly relevant for shaping core components of the system.

Stakeholder Expectation Analysis (Informal Discussions)

Informal discussions with people involved in local charitable activities provided valuable context about user expectations. Stakeholders emphasized the importance of clearer donation confirmations, improved campaign visibility, options for donor anonymity, and the secure storage of financial information. These insights offered a user-centered perspective that complemented findings from online observations and document analysis.

Requirements Categories:

- **Functional:** Specific actions the system must perform, such as campaign creation, donation logging, and donor management.
- **Non-functional:** System qualities including security, usability, performance, reliability, and data consistency.

2.3.2 Results of Requirement Gathering

The requirement-gathering activities revealed several major gaps in the current charity workflow, which directly shaped the functional and non-functional requirements of the proposed system.

Issue Identified	Description	Percentage of Respondents (%)
Lack of centralized record-keeping	Financial data is maintained using handwritten notes or scattered digital formats, making verification and reporting difficult	82%
Inefficient administrative oversight	Difficulty in supervising charity operations and generating consolidated financial summaries	79%
Absence of structured campaign visibility	Donors lack clear and consistent information on active campaigns and their progress	76%
Need for donor anonymity	Donors prefer privacy when making contributions and avoiding public disclosure	48%

Table 2.1 Summary of Requirement Gathering Survey Results

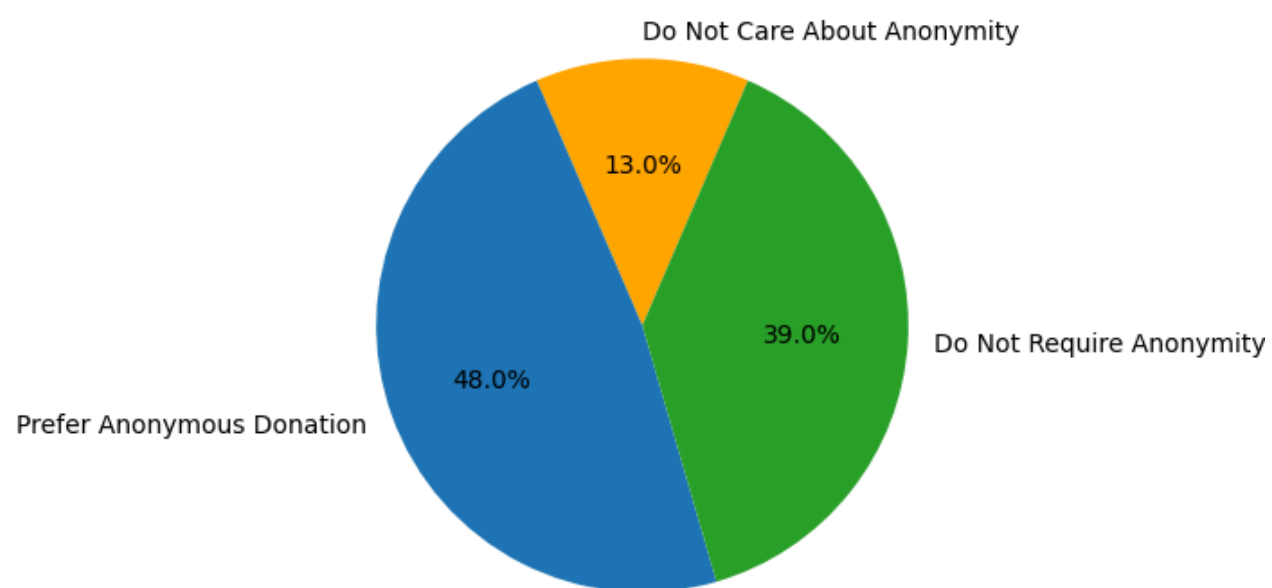


Figure 2.1 Donor preferences regarding anonymity

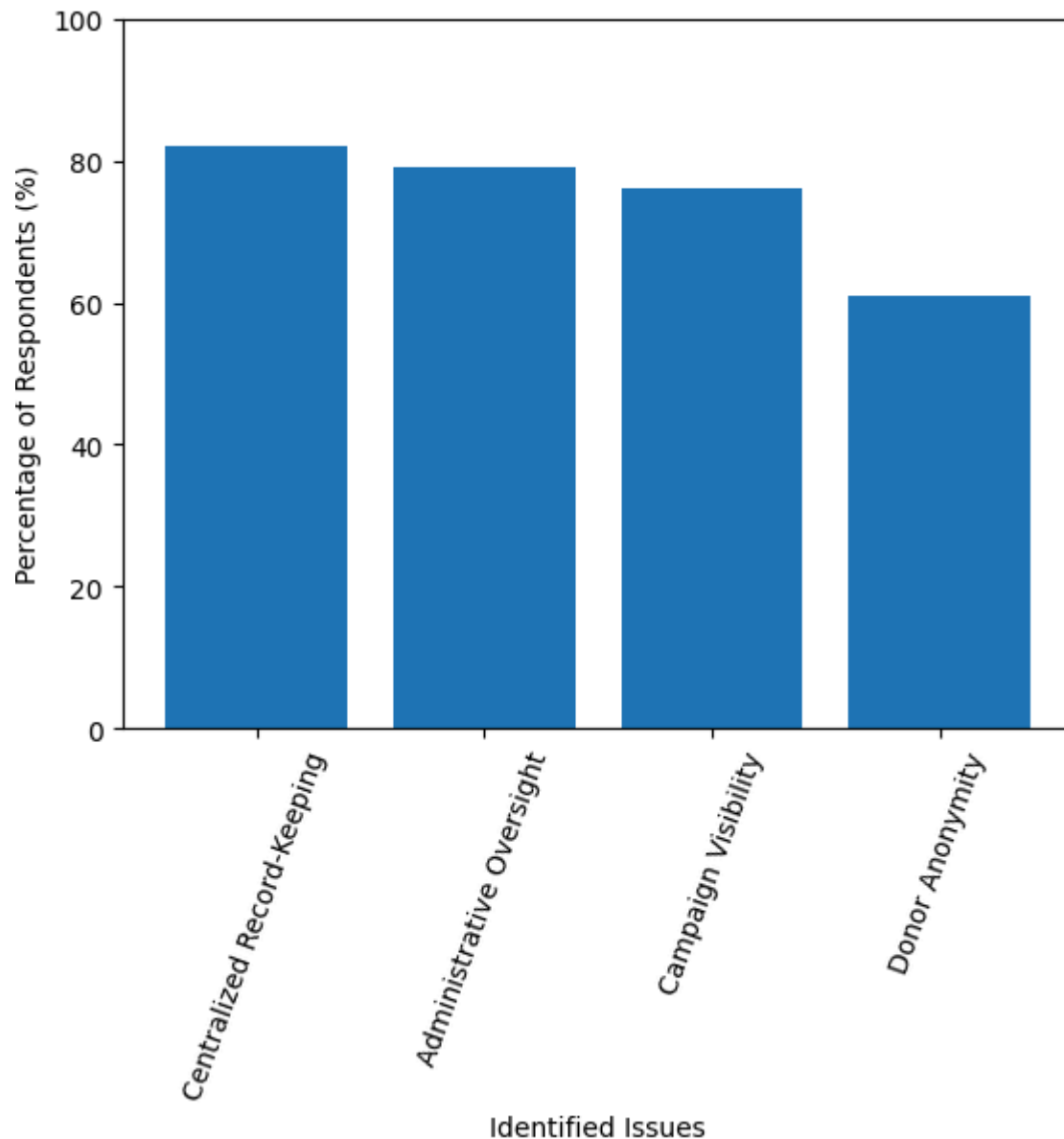


Figure 2.2 Survey Results: Major Issues in Existing Charity Workflow

2.4 Proposed System

2.4.1 Overview

The proposed charity management system is an integrated, web based platform designed to simplify, centralize, and digitize all charity-related operations. It connects donors, charity organizations, and administrators through a secure, transparent, and user-friendly interface. The system supports creating and managing campaigns, processing donations, tracking beneficiary allocations, and ensuring accountability through clear and accessible

records. By replacing the traditional manual process with automated workflows, real time updates, role-based access control, and data-driven reporting, the platform functions as a centralized hub where verified charities can present campaigns, donors can contribute securely and optionally anonymously, and administrators can supervise all activities.

2.4.2 Functional Requirements

A. Common Functional Requirements

FR-1: User Registration

The system must allow users of all roles donor, charity, and admin to securely register, log in, log out, and reset passwords.

FR-2: Search Campaigns

The system shall allow users to search for charity campaigns using keywords such as campaign title, charity name, or campaign description, enabling users to quickly locate specific campaigns that match their interests or intended causes.

FR-3: Filter Campaigns

The system shall allow users to filter available charity campaigns based on predefined criteria such as category, location, and campaign status (e.g., active or closed), helping users narrow down search results and easily identify campaigns that meet their preferences

FR-4: Notifications

The system must send email or in-app notifications for key events such as donation confirmations, campaign approvals, and system updates. This keeps users informed and ensures they receive timely updates relevant to their activities.

B. Donor Functional Requirements

FR-5: View Active Campaigns

The donor must be able to browse all approved and active campaigns. Campaign details including description, goal amount, progress, and charity information are visible to help donors make informed decisions.

FR-6: Make Donations

Donors must be able to donate securely through integrated payment gateways. The system must support optional anonymity, ensuring that donor identity is hidden from the charity when chosen.

FR-7: View Donation History

Donors must have access to a complete history of their donations, including timestamps, amounts, and associated campaigns. They must also be able to download official donation receipts for recordkeeping or tax purposes.

FR-8: Follow up Campaigns

Donors can choose to follow specific campaigns to receive updates such as progress notifications, closure announcements, or milestone achievements. This encourages ongoing engagement with causes they care about.

C. Charity Organization Functional Requirements**FR-9: Create Campaign**

Charity organizations must be able to submit new campaigns for admin approval. Campaign details include title, description, target amount, documents, and optional media. Submission triggers a workflow requiring administrative review.

FR-10: Edit Campaign

Charities can update certain campaign details, such as goal amount or campaign title. Every modification must be logged to maintain transparency and track changes over time.

FR-11: Close Campaign

Charities can close or complete campaigns once goals are met or when they choose to discontinue them. Closing stops new donations and updates the campaign's visibility status.

FR-12: View Donation Summary

Charities must be able to view summarized donation data related to their campaigns. This includes the total amount raised, number of donors, and basic information needed for reporting and internal planning.

FR-13: Receive Notifications

Charity organizations receive alerts for key events such as campaign approvals, new donations, or messages from the administrator. This ensures they remain updated and can manage their campaigns effectively.

D. Administrator Functional Requirements**FR-14: User Management**

Administrators must review and approve charity registrations, manage all user accounts, and handle issues such as disputes or policy violations. They hold the highest level of system privileges to maintain platform integrity.

FR-15: Donation Log Review

Admins must be able to view and review donation logs for compliance and transparency. This includes the ability to inspect transaction histories, and confirm donation legitimacy.

FR-16: System Settings

Administrators can configure global system settings such as campaign categories, platform rules, and other parameters that affect the overall operation of the platform. This ensures adaptability as the system grows.

2.4.3 Non-Functional Requirements**NFR-1. Security**

All user data must be fully protected from unauthorized access and cyber threats. Sensitive information including passwords, donor identities, and payment details must be encrypted both in transit using HTTPS/SSL and at rest. Role-based access control should ensure that only authorized users can access restricted areas, and secure authentication methods such as strong password policies and token-based sessions must be enforced. Additionally, the system must be safeguarded against common cybersecurity attacks to maintain overall protection.

NFR-2. Data Integrity

All stored data must remain accurate, consistent, and trustworthy. Unauthorized modifications to donation logs, campaign details, or beneficiary information must be prevented through strict controls. Validation rules must ensure that all entered data is correct and complete, while audit logs should record all important changes along with the responsible user and timestamp. Transaction management must prevent partial or corrupted entries, particularly during payment processes, to preserve data accuracy.

NFR-3. Reliability

The system must operate smoothly with minimal downtime or disruptions. It should remain functional even during sudden spikes in traffic or partial system failures. Regular backups must be performed to prevent data loss, and automatic recovery mechanisms such as failover systems should restore service quickly when issues occur. Critical operations like donations must never fail silently; errors should be logged, monitored, and retried whenever possible to ensure reliability.

NFR-4. Usability

The platform must be intuitive and easy to use for all users, regardless of their technical background. Interfaces should remain clean, simple, and consistent across all pages, while navigation must be straightforward enough to allow users to donate or create campaigns with minimal steps. Clear instructions, tooltips, and feedback messages should guide users throughout the process, and the UI must support accessibility needs through basic WCAG compliance, including readable fonts, proper contrast, and keyboard navigation.

NFR-5. Performance

The system must provide high responsiveness and fast loading times across all pages and user actions. Pages should load quickly even during peak donation periods with heavy traffic, and optimized database queries must be used to avoid delays when retrieving campaign or donation data. Caching frequently accessed information should reduce repeated processing, and APIs must respond within acceptable time limits to ensure a smooth and seamless user experience.

NFR-6. Scalability

The architecture must support growing numbers of users, campaigns, and donation volumes. The system should scale without performance degradation as more charities and donors join the platform. Both horizontal scaling (adding servers) and vertical scaling (upgrading server capacity) should be possible, and the database must be able to handle increasing records efficiently as the platform expands.

NFR-7. Maintainability

The system should be easy to update, debug, and extend over time. Code should follow modular, clean architecture principles so that new features can be added without breaking existing functionality. Developers must be able to understand and modify each module with minimal effort, supported by proper documentation such as API references, comments, and coding standards. Automated testing should also be in place to detect issues early and ensure stable deployments.

NFR-8. Compatibility

The platform must function correctly across different devices and technologies. It should work smoothly on all major browsers including Chrome, Firefox, Safari, and Edge, and

the UI must adapt effectively to various screen sizes such as laptops, tablets, and mobile phones.

2.4.4 Constraints / Pseudo Requirements

PR-1: Verified Charities

The system shall restrict participation to only registered and verified charity organizations. This constraint requires the platform to enforce a manual verification and approval process, where charities must submit legal and identification documents that are reviewed by system administrators before activation. As a result, charity onboarding cannot be fully automated and depends on external validation procedures, which may delay availability and limit rapid scaling.

PR-2: Legal and Regulatory Compliance

The system must comply with applicable data protection, financial, and charitable regulations within the operating region. This constraint limits how user and donation data can be collected, stored, processed, and retained. For example, personal data may require explicit user consent, sensitive information must be encrypted, and data retention periods may be legally fixed. The system cannot implement features that conflict with these regulatory requirements, even if they are technically feasible.

2.5 System Model (Overview)

The system model of the charity management system describes how different users interact with the system to achieve their goals. It provides a structured representation of the system's functional behavior from the user's perspective. This model focuses on identifying key actors, their interactions, and the services offered by the system.

The system model is represented using **scenarios and use case models**, which help illustrate real-world usage of the system in a clear and understandable manner. Scenarios explain specific sequences of actions performed by users, while use case models present the overall functionality and relationships between actors and the system. Together, these models help ensure that system requirements are clearly understood and provide a foundation for system design and implementation.

2.5.1 Scenario

A scenario represents a specific instance of a use case, describing a concrete sequence of actions that fulfills a particular goal within the system. Scenarios offer detailed insights into how users interact with the system, highlighting key processes and their realizations.

Scenario ID	SC-01
Scenario Name	User Login
Main Flow	1. The user navigates to the login page. 2. The user enters their email and password. 3. The system verifies the credentials. 4. Upon successful authentication, the system grants access and redirects the user to the appropriate dashboard.

Table 2.2 User Login Scenario

Scenario ID	SC-02
Scenario Name	Search Campaigns
Main Flow	1. The user navigates to the campaign listing or search page. 2. The user enters keywords or selects search criteria. 3. The system searches the campaign database based on the given input. 4. The system displays matching campaign results to the user.

Table 2.3 Search Campaigns Scenario

Scenario ID	SC-03
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Scenario Name	View Campaign Details
Main Flow	1. The user selects a campaign from the campaign list or search results. 2. The system retrieves the campaign details. 3. The system displays the full campaign information to the user.

Table 2.4 View Campaign Details Scenario

Scenario ID	SC-04
Scenario Name	Make Donation
Main Flow	1. The donor accesses the system. 2. The donor browses or searches for active campaigns. 3. The donor selects a campaign to support. 4. The system displays detailed campaign information. 5. The donor enters the donation amount. 6. The donor chooses whether to donate anonymously or provide basic information. 7. The donor selects a payment method. 8. The system securely processes the payment. 9. The system records the donation and updates the campaign progress. 10. The system displays a donation confirmation message and sends a notification if contact details were provided

Table 2.5 Make Donation Scenario

Scenario ID	SC-05
Scenario Name	Create Campaign
Main Flow	1. The charity organization logs into the system. 2. The charity selects the option to create a new campaign. 3. The system displays the campaign creation form. 4. The charity enters campaign details such as title, description, target amount, category, and duration. 5. The charity uploads required supporting documents and optional media files. 6. The charity submits the campaign for approval. 7. The system validates the submitted information. 8. The system stores the campaign with a pending status. 9. The administrator is notified about the new campaign submission.

Table 2.6 Create Campaign Scenario

Scenario ID	SC-06
Scenario Name	Verify Charity Registration

Main Flow	1. Administrator navigates to the pending charity registrations list. 2. Administrator selects a charity registration to review. 3. Administrator verifies submitted details, documents, and compliance requirements. 4. The administrator approves or rejects the registration. 5. The system updates the charity's account status and notifies the charity organization.
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Table 2.7 Verify Charity Registration Scenario

2.5.2 Use Case Model

A use-case model is a model of how different types of users interact with the system. Use cases describe the typical interactions between the users of a system and the system itself, providing a brief narrative of how a system is used.

2.5.2.1 Use Case Diagram

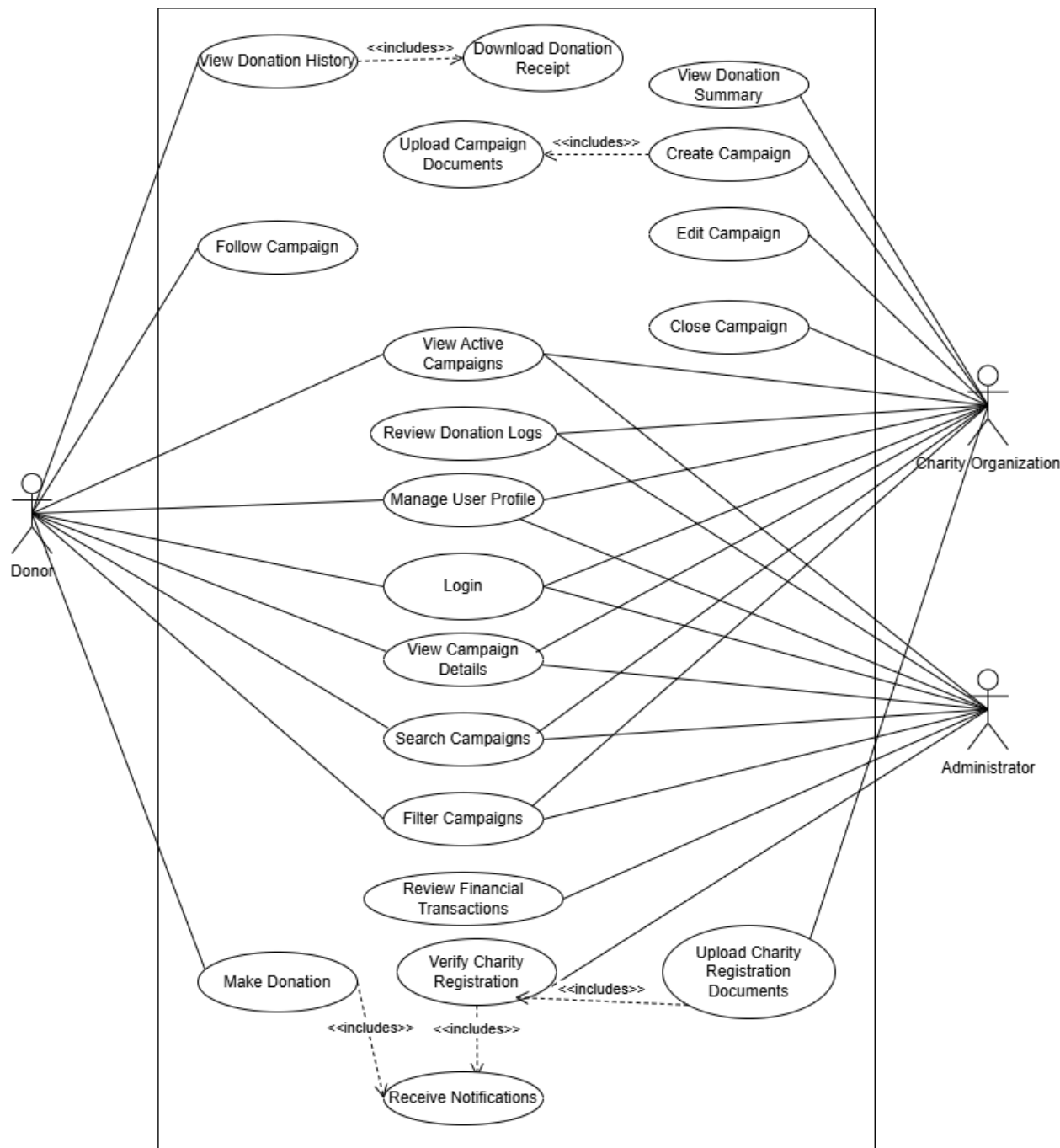


Figure 2.3 Use case diagram

2.5.2.2 Description of Use-Case Model

Use Case ID	UC-01
-------------	-------

Use Case Name	Login
Participating Actors	Donor, Charity Organization, Administrator
Use Case Description	Allows registered users to authenticate into the system using valid credentials. If the user forgets their password, the system provides a password reset option as part of the login process.
Precondition	The user is registered in the system and has a valid email and password.
Flow of Events	1. The user navigates to the login page. 2. The user enters email and password. 3. The system validates the credentials. 4. If valid, the system grants access based on the user's role. 5. Users are redirected to their respective dashboard.
Alternate Flow (Reset Password)	1. The user selects the "Forgot Password" option. 2. The system prompts the user to enter their registered email. 3. The system sends a password reset link or code. 4. The user sets a new password. 5. The user returns to the login page and logs in successfully.
Postconditions	The user is successfully authenticated and gains access to authorized system features.

Table 2.8 Login Use Case Description

Use Case ID	UC-02
-------------	-------

Use Case Name	Manage User Profile
Actor	Donor, Charity Organization, Administrator
Description	Allows a registered user to view and update their personal profile information.
Preconditions	User is logged in
Flow of Events	1. User accesses their profile page 2. System displays current profile information 3. User edits profile details (e.g., name, contact information, password) 4. User submits the changes 5. System validates and saves the updated information 6. System confirms successful update
Alternative Flows	Invalid or incomplete information entered - system displays error message User cancels profile update - no changes are saved
Postconditions	User profile information is updated successfully

Table 2.9 Manage User Profile Use Case Description

Use Case ID	UC-03
Use Case Name	View Active Campaigns
Participating Actors	Donor, Charity Organization, Adminstrator
Use Case Description	Allows users to see all campaigns that are currently active and open for donations. This helps donors choose campaigns to contribute to and charities monitor ongoing campaigns.

Precondition	The system has active campaigns stored in the database. User may or may not be logged in.
Flow of Events	1. The user navigates to the campaign list. 2. The system retrieves all active campaigns. 3. The system displays active campaigns with details (title, description, goal, progress, and charity info). 4. The user selects a campaign for further actions (e.g., view details, donate).
Alternate Flow	3a. If there are no active campaigns, the system displays a “No active campaigns found” message.
Post Condition	Users are presented with a current list of campaigns open for donations, ready for further interaction.

Table 2.10 View Active Campaigns Use Case Description

Use Case ID	UC-04
Use Case Name	Search Campaigns
Participating Actors	Donor, Charity Organization, Administrator
Use Case Description	Allows users to search for charity campaigns based on keywords or basic criteria in order to find relevant campaigns.
Precondition	The user has access to the system (login is optional).

Flow of Events	1. The user navigates to the campaign listing or search page. 2. The user enters search keywords or criteria. 3. The system processes the search request. 4. The system displays a list of matching campaigns.
Alternate Flow	If no matching campaigns are found, the system displays a “No campaigns found” message.
Postconditions	The user receives a list of campaigns matching the search criteria. No changes are made to the system data.

Table 2.11 Search Campaigns Use Case Description

Use Case ID	UC-05
Use Case Name	Filter Campaigns
Participating Actors	Donor, Charity Organization, Administrator
Use Case Description	Allows users to narrow down the list of campaigns based on selected criteria such as category, location, or campaign status to find relevant campaigns efficiently.
Precondition	The system is available and a list of campaigns exists. Users may or may not be logged in.
Flow of Events	1. The user opens the campaign list. 2. The user selects one or more filter options (e.g., category, location, status). 3. The system applies the selected filters. 4. The system displays the filtered list of campaigns.

Alternate Flow	If no campaigns match the selected criteria, the system displays a “No campaigns found” message..
Post Condition	The user views a filtered list of campaigns based on the selected criteria.

Table 2.12 Filter Campaigns Use Case Description

Use Case ID	UC-06
Use Case Name	View Campaign Details
Participating Actors	Donor, Charity Organization, Administrator
Use Case Description	Allows users to view detailed information about a selected charity campaign, including description, goal amount, progress, and status.
Precondition	The campaign exists in the system.
Flow of Events	1. The user selects a campaign from the campaign list or search results. 2. The system retrieves the campaign details. 3. The system displays the full campaign information to the user.
Postconditions	The user has viewed the campaign details. No changes are made to the campaign data

Table 2.13 View Campaign Details Use Case Description

Use Case ID	UC-07
Use Case Name	Receive Notifications

Participating Actors	Donor, Charity Organization, Administrator
Use Case Description	Allows users to receive real-time alerts and messages from the system regarding important events such as donations, campaign updates, and system announcements.
Precondition	The user is registered in the system (optional: logged in for in-app notifications). Notifications are enabled in the system.
Flow of Events	1. An event occurs in the system (e.g., a donation is made, campaign status changes, or system update). 2. The system generates a notification. 3. The system delivers the notification to the relevant user(s) via email or in-app message. 4. The user views the notification.
Alternate Flow	3a. If the notification cannot be delivered (e.g., email failure), the system logs the event for later delivery or retries.
Post Condition	Users are informed about relevant events, enabling timely actions and awareness of system updates.

Table 2.14 Receive Notifications Use Case Description

Use Case ID	UC-08
Use Case Name	Make Donation
Actor	Donor (Registered User or Guest)
Description	Allows a donor to contribute funds to a selected charity campaign.

Preconditions	Campaign exists and is active
Flow of Events	1. Donor selects a campaign 2. Donor enters donation amount 3. Donor proceed to payment 4. Donor confirms the donation 5. System processes the payment 6. System records the donation 7. System displays donation confirmation
Alternative Flows	Donor selects anonymous donation Payment fails - system displays error and donation is not recorded Donor cancels donation before confirmation
Postconditions	Donation is stored Campaign total amount is updated

Table 2.15 Make Donation Use Case Description

Use Case ID	UC-9
Use Case Name	View Donation History
Actor	Donor (Registered User)
Description	Allows a donor to view their past donation records.
Preconditions	User is logged in
Flow of Events	1. User accesses donation history 2. System displays list of past donations
Alternative Flows	User downloads donation receipt No donation records - system displays message

Postconditions	Donation history is displayed Receipt may be downloaded
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Table 2.16 View Donation History Use Case Description

Use Case ID	UC-10
Use Case Name	Download Donation Receipt
Participating Actors	Donor(Registered User)
Use Case Description	Allows donors to download official receipts for their donations. Receipts can be used for personal records or tax purposes.
Precondition	Donor has completed at least one donation and is logged in to the system. The system has recorded the transaction.
Flow of Events	1. Donor navigates to their donation history. 2. Donor selects a completed donation. 3. The system generates the official receipt in a downloadable format (e.g., PDF). 4. Donor downloads or saves the receipt.
Alternate Flow	3a. If the receipt cannot be generated due to a system error, the system displays an error message and logs the issue.
Post Condition	Donor successfully obtains the official receipt for recordkeeping or tax purposes.

Table 2.17 Download Donation Receipt Use Case Description

Use Case ID	UC-11
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Use Case Name	Upload Charity Registration Document
Participating Actors	Charity Organization
Use Case Description	Allows a charity organization to upload required legal and registration documents during the registration process. These documents are reviewed by the administrator to verify the legitimacy of the charity before account approval.
Preconditions	- Charity organization is registering or updating registration details- User has access to the registration form
Flow of Events	1. Charity organization initiates registration or verification process 2. System displays document upload section 3. Charity uploads required registration documents (e.g., license, legal certificate) 4. System validates file type and size 5. System stores documents securely and marks registration as pending review
Alternate Flow	4a. Invalid or unsupported document format – system displays error message 3a. Charity skips upload – system prevents submission and displays warning
Postconditions	- Registration documents are stored in the system- Charity registration status is set to “Pending Approval”- Administrator is notified for verification

Table 2.18 Upload Charity Registration Document Use Case Description

Use Case ID	UC-12
Use Case Name	Upload Campaign Document

Participating Actors	Charity Organization
Use Case Description	Allows a charity organization to upload supporting documents required for a campaign before or during campaign creation. These documents help validate the legitimacy and purpose of the campaign and are reviewed by the administrator as part of the campaign approval process.
Preconditions	- Charity organization is registered and logged into the system - Charity has initiated the campaign creation process
Flow of Events	1. Charity organization navigates to the “Create Campaign” page 2. System displays campaign creation form including document upload section 3. Charity uploads required campaign documents (e.g., approval letters, project plans, licenses) 4. System validates document type and size 5. System securely stores uploaded documents and links them to the campaign 6. Charity completes and submits the campaign for approval
Alternate Flow	3a. Charity skips required document upload – system prevents submission and displays warning 4a. Unsupported or invalid document format – system displays an error message and requests re-upload
Postconditions	-Campaign documents are successfully stored in the system- Documents are associated with the submitted campaign- Campaign status remains Pending Approval until reviewed by administrator

Table 2.19 Upload Campaign Document Use Case Description

Use Case ID	UC-13
Use Case Name	Create Campaign
Participating Actors	Charity Organization
Use Case Description	Allows a charity organization to create a new donation campaign by providing details such as campaign title, description, goal amount, and optional media or supporting documents.
Precondition	Charity organizations are registered and logged in to the system.
Flow of Events	1. Charity organization navigates to the “Create Campaign” page. 2. Charity fills in campaign details (title, description, goal amount, documents, media). 3. Charity submits the campaign. 4. The system saves the campaign and displays it in the charity’s campaign list.
Alternate Flow	3a. If required fields are missing or invalid, the system shows an error message and prompts for correction.
Post Condition	A new campaign is created and stored in the system, ready to receive donations.

Table 2.20 Create Campaign Use Case Description

Use Case ID	UC-14
Use Case Name	Edit Campaigns
Participating Actors	Charity Organization
Use Case Description	Allows a charity organization to modify existing campaign information such as title, goal amount.
Precondition	The charity organization is logged into the system and owns the campaign being edited.

Flow of Events	1. The charity organization navigates to its list of campaigns. 2. The charity organization selects a campaign to edit. 3. The system displays the campaign's current details. 4. The charity organization updates the desired fields. 5. The system validates and saves the changes. 6. The system confirms the successful update.
Alternate Flow	If the charity organization tries to edit a campaign that is already closed or locked by the system, the edit action is not allowed. The system informs the user that the campaign cannot be modified, and no changes are saved.
Postconditions	The campaign's information is updated in the system. Changes are reflected in all related views (campaign listings, search results, donation logs). A system log entry is created to record the update (optional but recommended).

Table 2.21 Edit Campaigns Use Case Description

Use Case ID	UC-15
Use Case Name	Close Campaign
Actor	Charity Organization
Description	Allows a charity organization to close an active campaign when fundraising ends.
Preconditions	- Campaign exists - Campaign is active - User is authorized owner

Flow of Events	1. Charity selects an active campaign 2. Charity chooses “Close Campaign” 3. System marks campaign as closed 4. System disables further donations 5. System sends notifications to relevant users
Alternative Flows	Campaign already closed - message displayed Unauthorized access - action denied
Postconditions	Campaign status updated Relevant users notified

Table 2.22 Close Campaign Use Case Description

Use Case ID	UC-16
Use Case Name	View Donation Summary
Actor	Charity Organization
Description	Allows a charity organization to view summarized donation information for their campaigns.
Preconditions	- User is logged in - User owns at least one campaign
Flow of Events	1. Charity selects a campaign 2. System displays total donations and summary statistics
Alternative Flows	No donations received - empty summary shown Campaign not found - error message
Postconditions	Donation summary is displayed

Table 2.23 View Donation Summary Use Case Description

Use Case ID	UC-17
Use Case Name	Verify Charity Registration
Participating Actors	Administrator
Use Case Description	Allows the administrator to review and verify newly registered charity organizations to ensure they meet eligibility criteria and comply with platform policies. This ensures only legitimate charities can create campaigns and receive donations.
Precondition	The charity organization has submitted a registration request and the administrator is logged in to the system.
Flow of Events	1. Administrator navigates to the pending charity registrations list. 2. Administrator selects a charity registration to review. 3. Administrator verifies submitted details, documents, and compliance requirements. 4. The administrator approves or rejects the registration. 5. The system updates the charity's account status and notifies the charity organization.
Alternate Flow	4a. If the registration is incomplete or invalid, the administrator rejects the registration and may send feedback for correction.
Post Condition	Charity registration is either approved (account activated) or rejected (account remains inactive), ensuring only verified charities participate on the platform.

Table 2.24 Verify Charity Registration Use Case Description

Use Case ID	UC-18
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Use Case Name	Review Donation Logs
Participating Actors	Charity Organization, Administrator
Use Case Description	Allows a charity organization to review donation records related to its campaigns, and allows the administrator to monitor all donation activities for reviewing and system oversight purposes.
Precondition	The user is logged into the system and has appropriate access privileges.
Flow of Events	1. The user navigates to the donation logs section. 2. The system verifies the user's role and access privileges. 3. If the user is a Charity Organization: • The system retrieves donation records related only to the organization's campaigns. 4. If the user is an Administrator: • The system retrieves all donation records across the platform for monitoring purposes. 5. The system displays the donation logs in a structured format (e.g., donor name, amount, date, campaign). 6. The user reviews the displayed donation information.
Alternate Flow	If no donation records are found, the system displays an appropriate message.
Postconditions	The user has reviewed donation records appropriate to their role. No donation records are modified during review. System logs the access for reviewing purposes (especially for administrators).

Table 2.25 Review Donation Logs Use Case Description

Use Case ID	UC-19
Use Case Name	Review Financial Transactions
Participating Actors	Administrator
Use Case Description	Allows the administrator to review donation logs and monitor all financial transactions in the system to ensure transparency and maintain system oversight. This use case is limited to viewing transaction records only and does not permit modification.
Precondition	The administrator is logged into the system. The administrator has permission to view transaction records.
Flow of Events	1. The administrator navigates to the transaction review logs section. 2. The system retrieves transaction records from the database. 3. The system displays transaction details such as donor (or anonymous), campaign, donated amount, date, and transaction status. 4. The administrator reviews the transaction records. 5. The administrator may mark transactions for further review if irregularities are noticed. 6. The system records that the transactions were viewed by the administrator.
Alternate Flow	If the system cannot retrieve the financial transactions due to a database error or connectivity issue, it displays an error message and prompts the administrator to try again later.
Postconditions	The administrator has reviewed the available financial transactions. Any transactions marked for further review are recorded by the system. No transaction data is modified. Transaction records remain unchanged and intact.

Table 2.26 Review Financial Transactions Use Case Description

Use Case ID	UC-20
Use Case Name	Follow Campaign
Participating Actors	Donor
Use Case Description	This use case allows a donor to follow a specific charity campaign in order to receive updates such as donation milestones, campaign progress, and closure notifications, permit modification.
Precondition	The donor must be logged in and have access to the campaign listing or campaign details page.
Flow of Events	1. Donor selects a campaign to follow. 2. Donor clicks the “Follow” button. 3. The system records the donor as a follower of the campaign. 4. The system confirms that the donor is now following the campaign.
Alternate Flow	If the donor is already following the campaign, the system notifies them that they are already a follower and prevents duplicate entries.
Postconditions	The donor is added to the campaign’s follower list and will receive notifications for updates related to that campaign.

Table 2.27 Follow Campaign Use Case Description

2.5.3 Object Model

2.5.3.1 Data Dictionary

Field	Type	Length	Description	Example
user_id	INT	6	Primary key, auto-incremented user identifier	000001
full_name	VARCHAR	100	User's full name	Abel Getachew
email	VARCHAR	100	User email address	abel@email.com
password_hash	VARCHAR	255	Encrypted password	\$2y\$10\$...
role	ENUM	–	User role (Donor, Charity, Admin)	Donor
status	ENUM	–	Account status (Active, Suspended)	Active
registration_date	DATETIME	–	Date of account registration	2025-03-20
last_login	DATETIME	–	Last login timestamp	2025-03-22

Table 2.28: User

Field	Type	Length	Description	Example
charity_id	INT	6	Primary key	101
user_id	INT	6	Foreign key to User	000002

organization_name	VARCHAR	150	Registered charity name	Help Ethiopia
registration_number	VARCHAR	50	Legal registration number	CH-88921
verification_status	ENUM	–	Pending, Verified, Rejected	Verified
verification_date	DATETIME	–	Date of verification	2025-03-21

Table 2.29: CharityOrganization

Field	Type	Length	Description	Example
campaign_id	INT	6	Primary key	201
charity_id	INT	6	Foreign key to CharityOrganization	101
title	VARCHAR	150	Campaign title	Feed Children
description	TEXT	–	Campaign description	Food aid program
target_amount	DECIMAL	–	Fundraising goal	50000
current_amount	DECIMAL	–	Amount raised	32000
status	ENUM	–	Active, Closed	Active
start_date	DATE	–	Campaign start date	2025-03-01
end_date	DATE	–	Campaign end date	2025-06-01

Table 2.30 Campaign

Field	Type	Length	Description	Example
donation_id	INT	6	Primary key	301

campaign_id	INT	6	Foreign key to Campaign	201
donor_id	INT	6	Foreign key to User	000003
amount	DECIMAL	–	Donation amount	500
donation_date	DATETIME	–	Date of donation	2025-03-22
is_anonymous	BOOLEAN	–	Anonymous donation flag	true

Table 2.31 Donation

Field	Type	Length	Description	Example
receipt_id	INT	6	Primary key	401
donation_id	INT	6	Foreign key to Donation	301
issued_date	DATETIME	–	Receipt issue date	2025-03-22
receipt_reference	VARCHAR	50	Receipt reference code	REC-90321

Table 2.32: DonationReceipt

Field	Type	Length	Description	Example
document_id	INT	6	Primary key	501
campaign_id	INT	6	Foreign key to Campaign	201
document_type	VARCHAR	50	Type of document	License
file_path	VARCHAR	255	Document file path	/docs/license.pdf
upload_date	DATETIME	–	Date uploaded	2025-03-18

Table 2.33 CampaignDocument

Field	Type	Length	Description	Example
follow_id	INT	6	Primary key	601
campaign_id	INT	6	Foreign key to Campaign	201
user_id	INT	6	Foreign key to User	000003
follow_date	DATETIME	–	Date followed	2025-03-19

Table 2.34: CampaignFollow

Field	Type	Length	Description	Example
notification_id	INT	6	Primary key	701
user_id	INT	6	Recipient user	000001
title	VARCHAR	100	Notification title	Campaign Closed
message	TEXT	–	Notification message	Fundraising ended
is_read	BOOLEAN	–	Read status	false
sent_date	DATETIME	–	Date sent	2025-03-23

Table 2.35 Notification

Field	Type	Length	Description	Example
log_id	INT	6	Primary key	801
admin_id	INT	6	Foreign key to User (Admin)	000010
action_type	VARCHAR	100	Type of action	Verify Charity
action_date	DATETIME	–	Date of action	2025-03-24
description	TEXT	–	Action description	Charity approved

Table 2.36 ReviewLog

Field	Type	Length	Description	Example
verification_id	INT	6	Primary key y	901
charity_id	INT	6	Foreign key to CharityOrganization	101
admin_id	INT	6	Admin who verified	000010
status	ENUM	–	Approved, Rejected	Approved
remarks	TEXT	–	Verification remarks	Valid documents
verification_date	DATETIME	–	Date verified	2025-03-24

Table 2.37 CharityVerification

Field	Type	Length	Description	Example
reset_id	INT	6	Primary key	1001

user_id	INT	6	Foreign key to User	000001
reset_code	CHAR	6	Password reset code	458392
expiration_date	DATETIME	–	Code expiration time	2025-03-25

Table 2.38 PasswordReset

2.5.3.2 Class Modeling

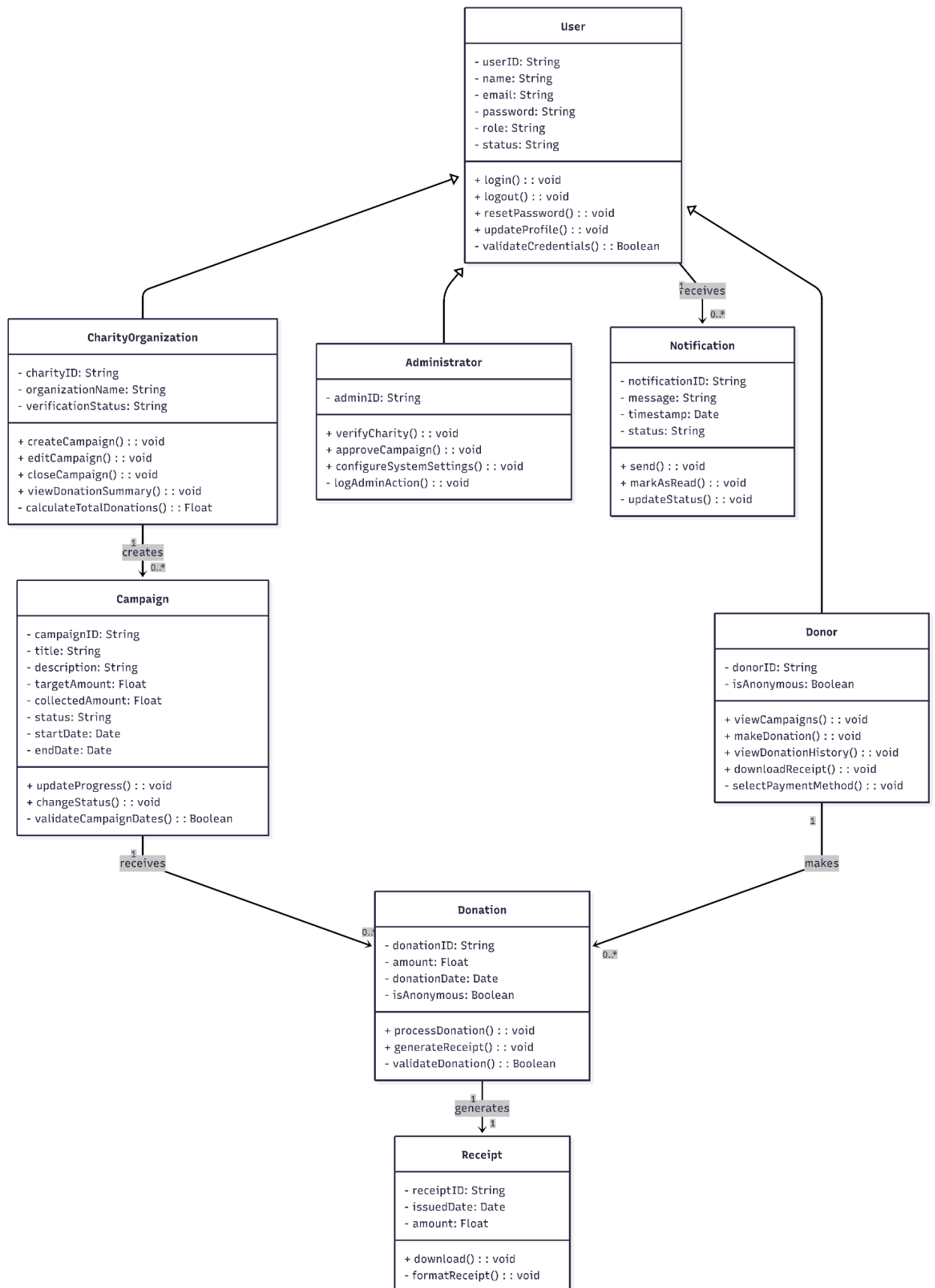


Figure 2.4 Class diagram

2.5.3.3 Dynamic Modeling

Sequence diagrams

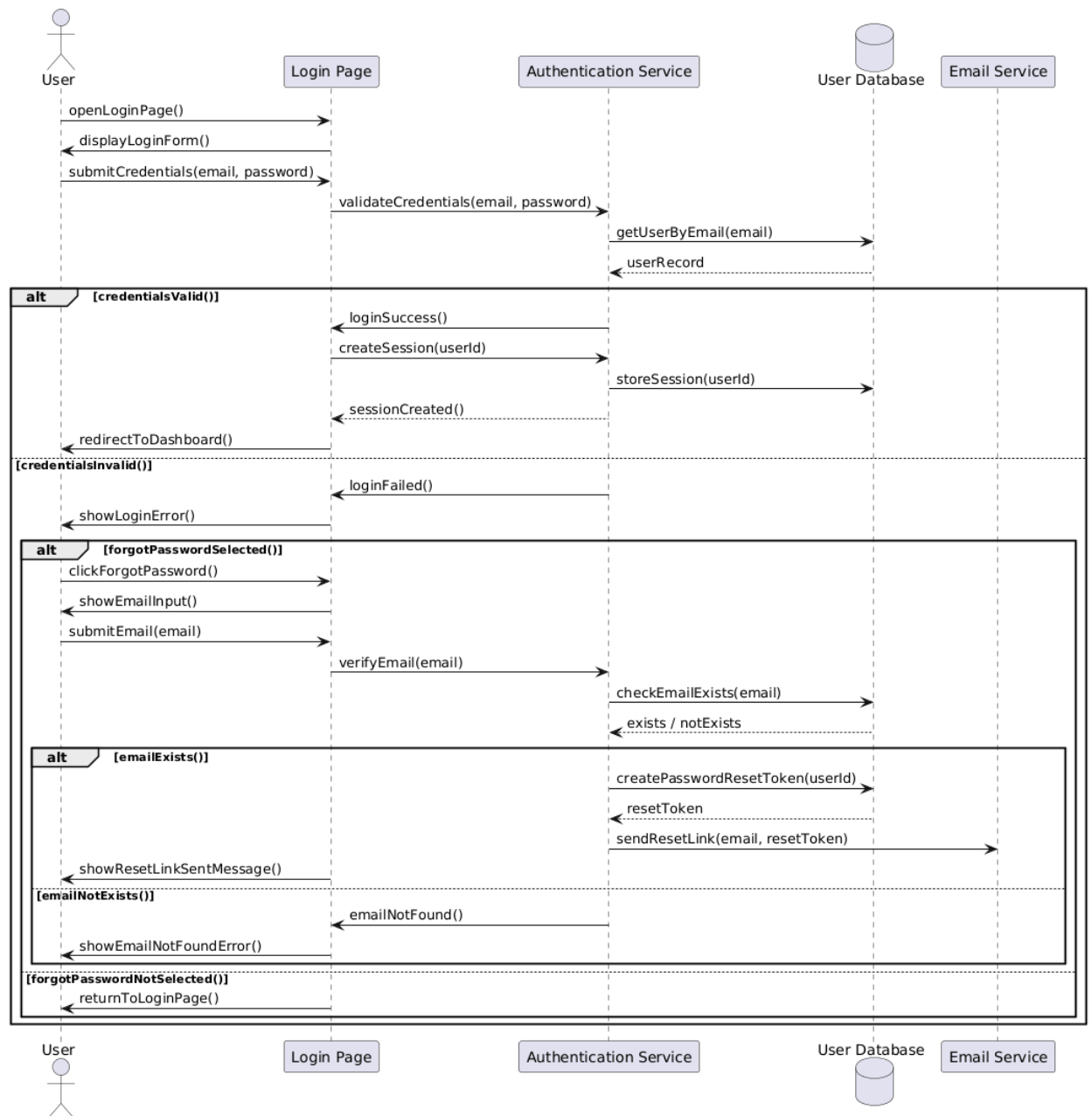


Figure 2.5 Login Sequence diagram with reset password

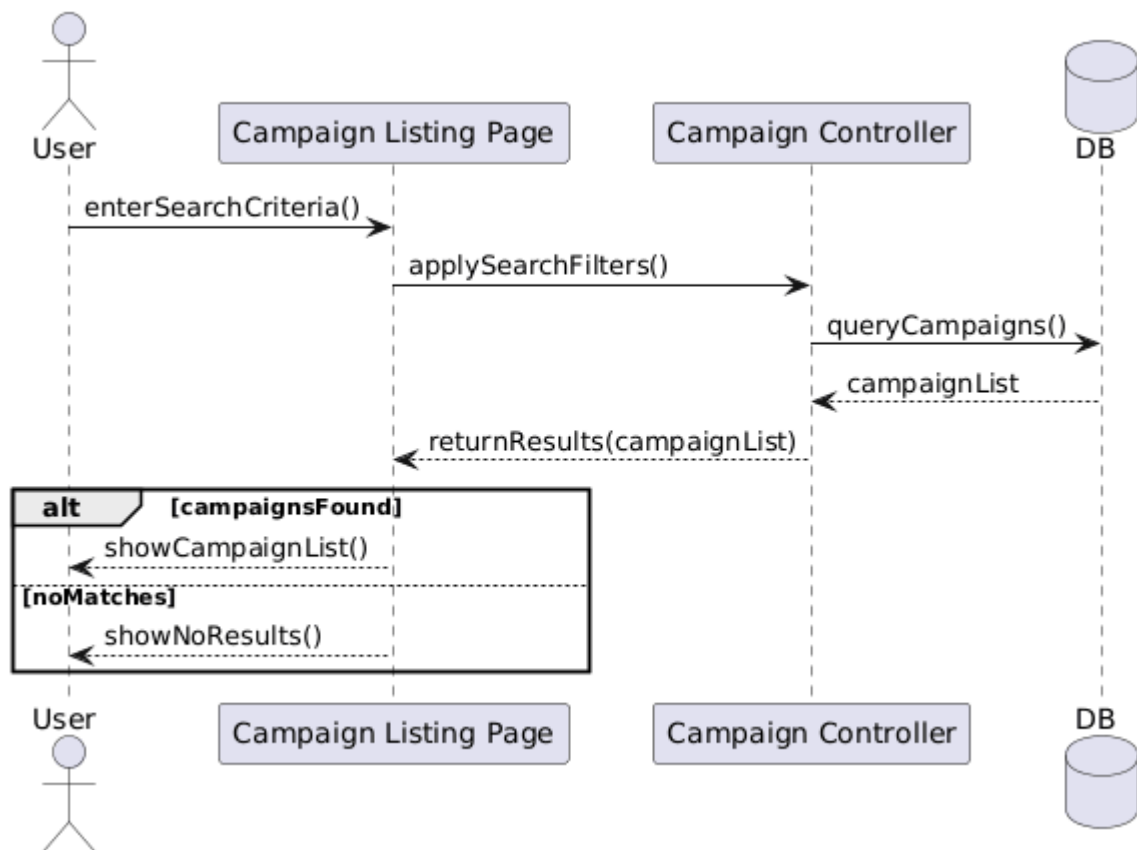


Figure 2.6 Search and filter campaigns sequence diagram

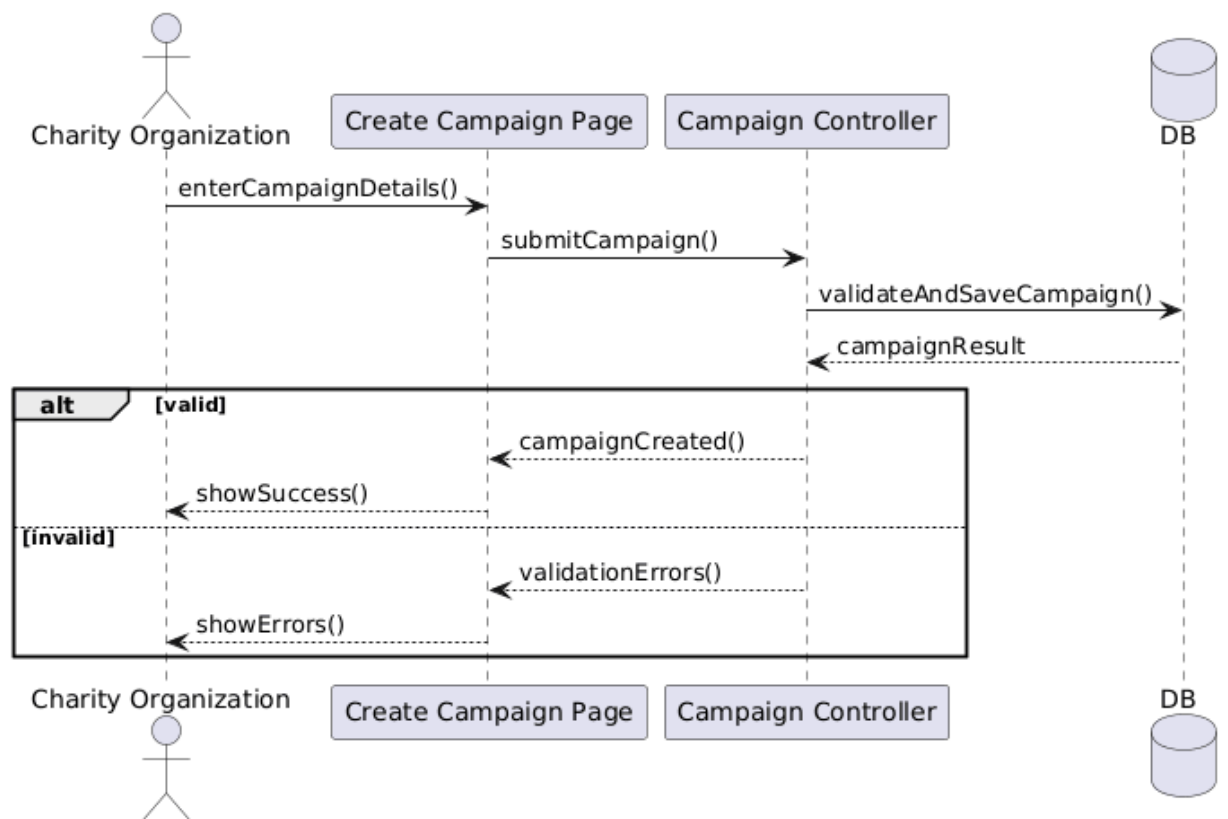


Figure 2.7 Create campaign sequence diagram

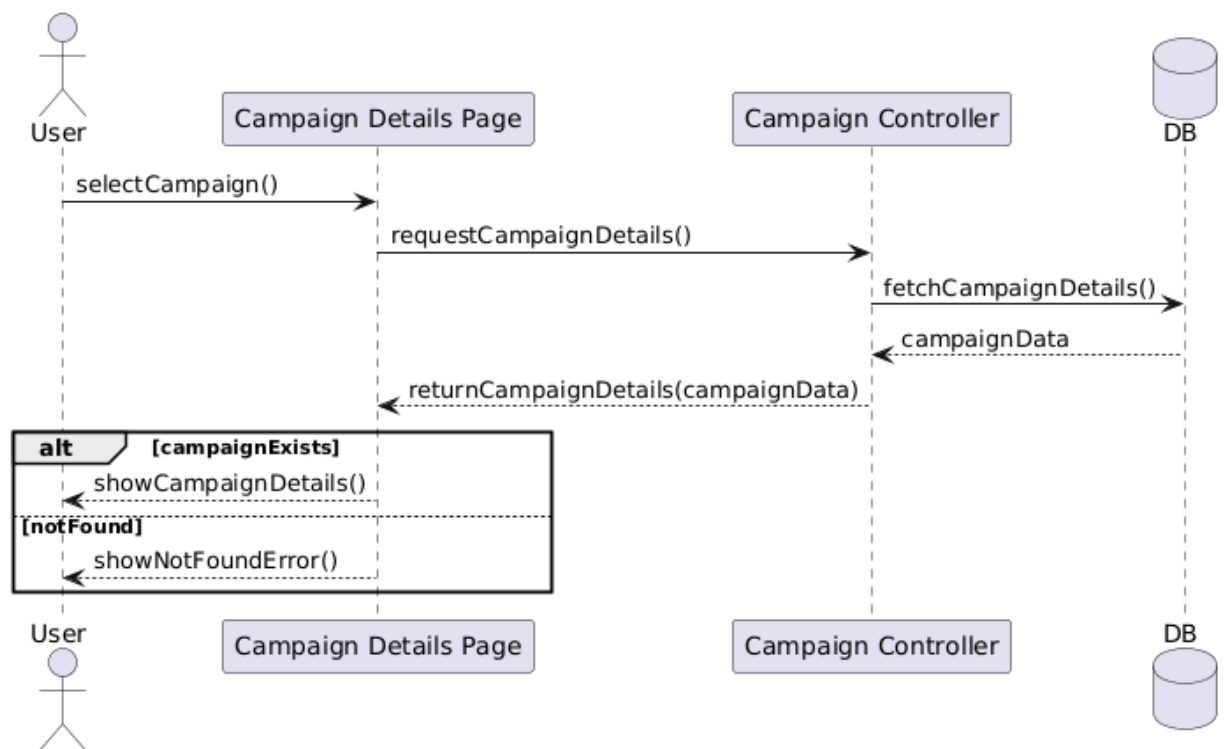


Figure 2.8 View campaign details sequence diagram

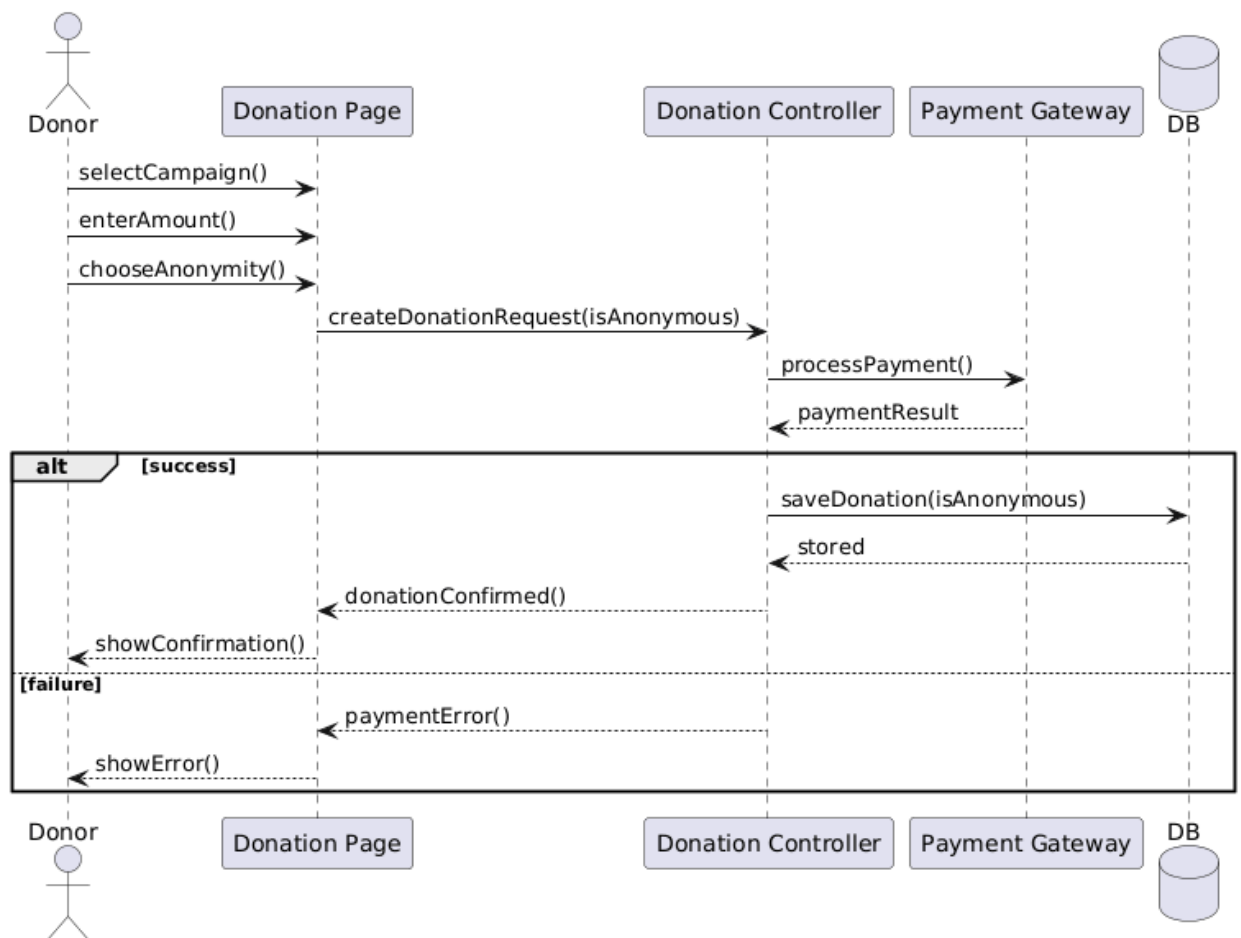


Figure 2.9 Make donation sequence diagram

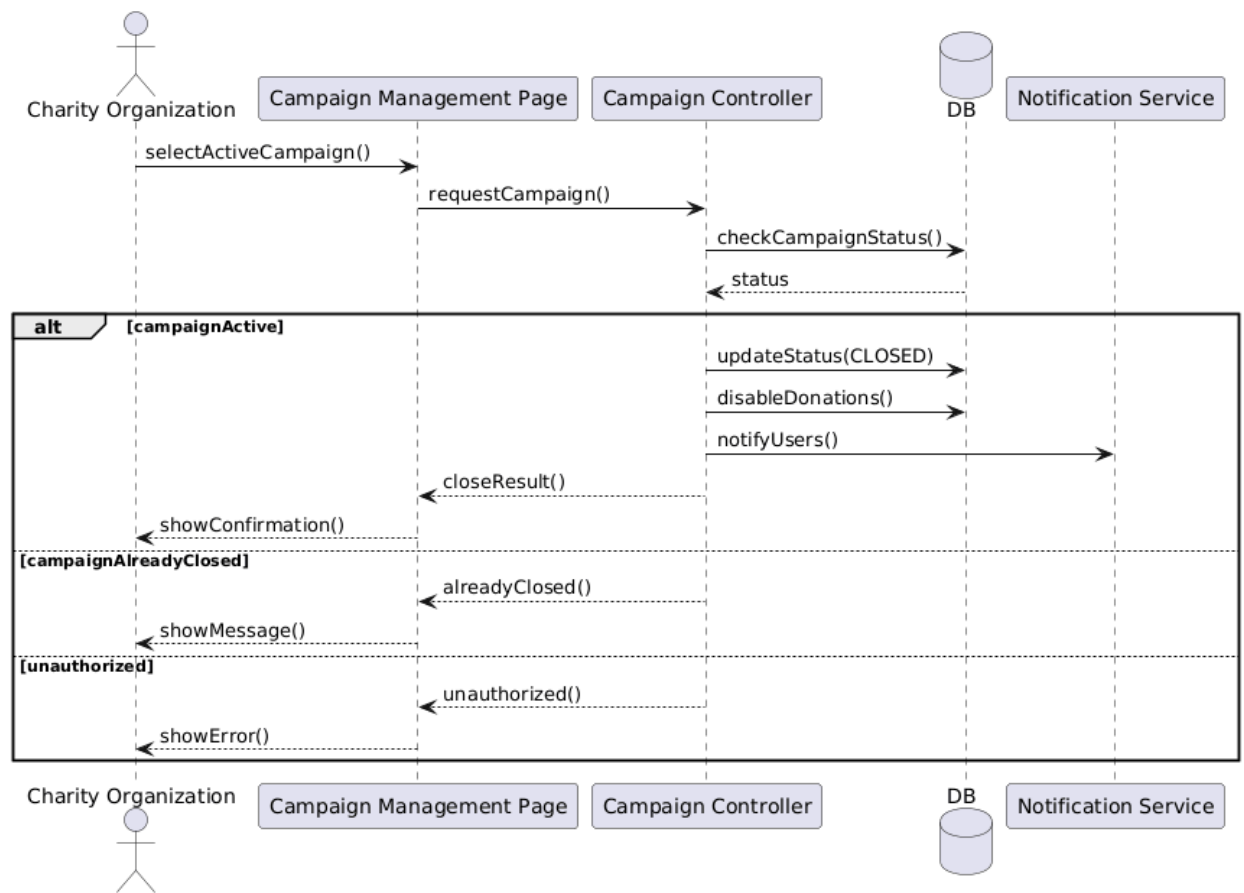


Figure 2.10 Close campaign sequence diagram

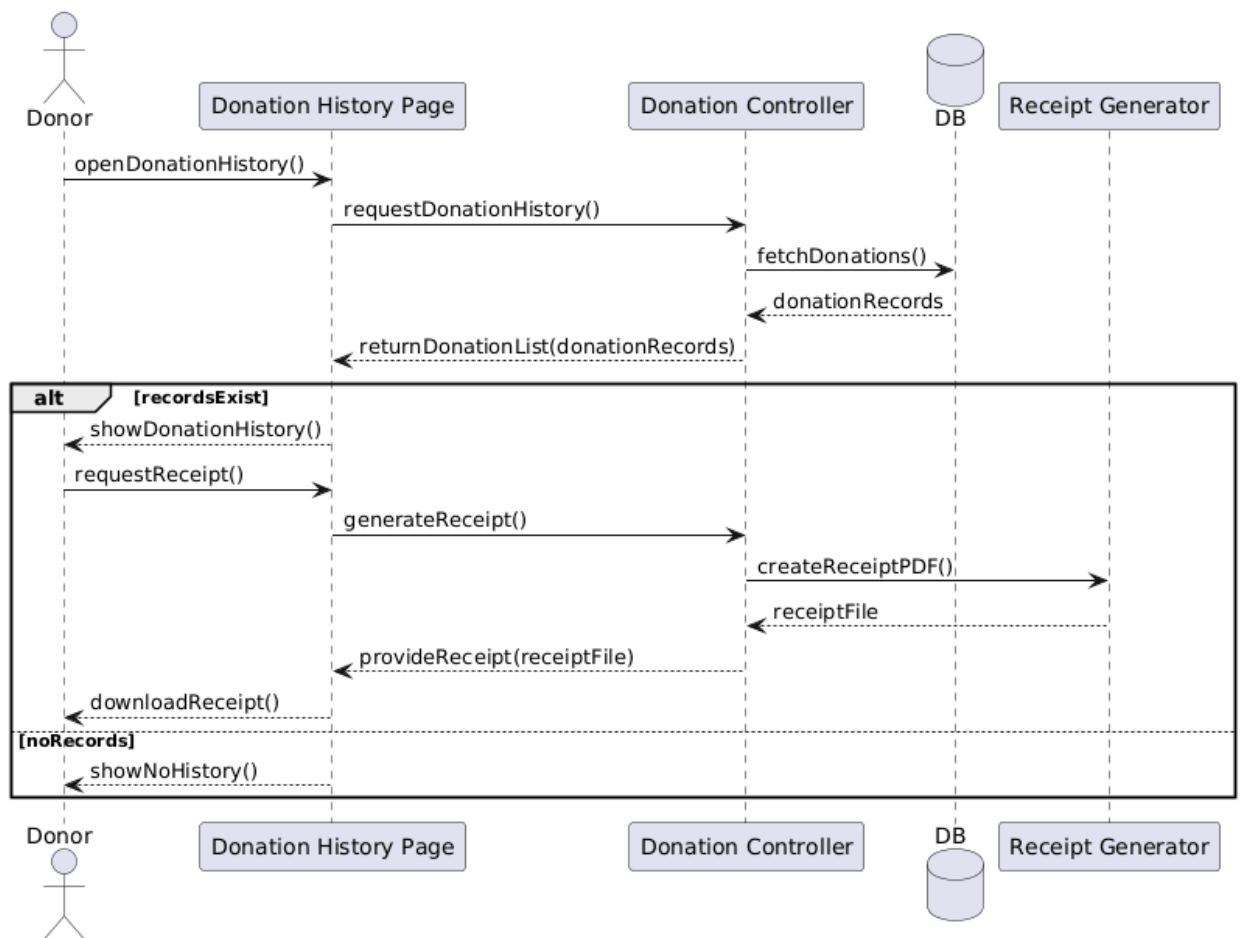


Figure 2.11 View donation history and download receipt sequence diagram

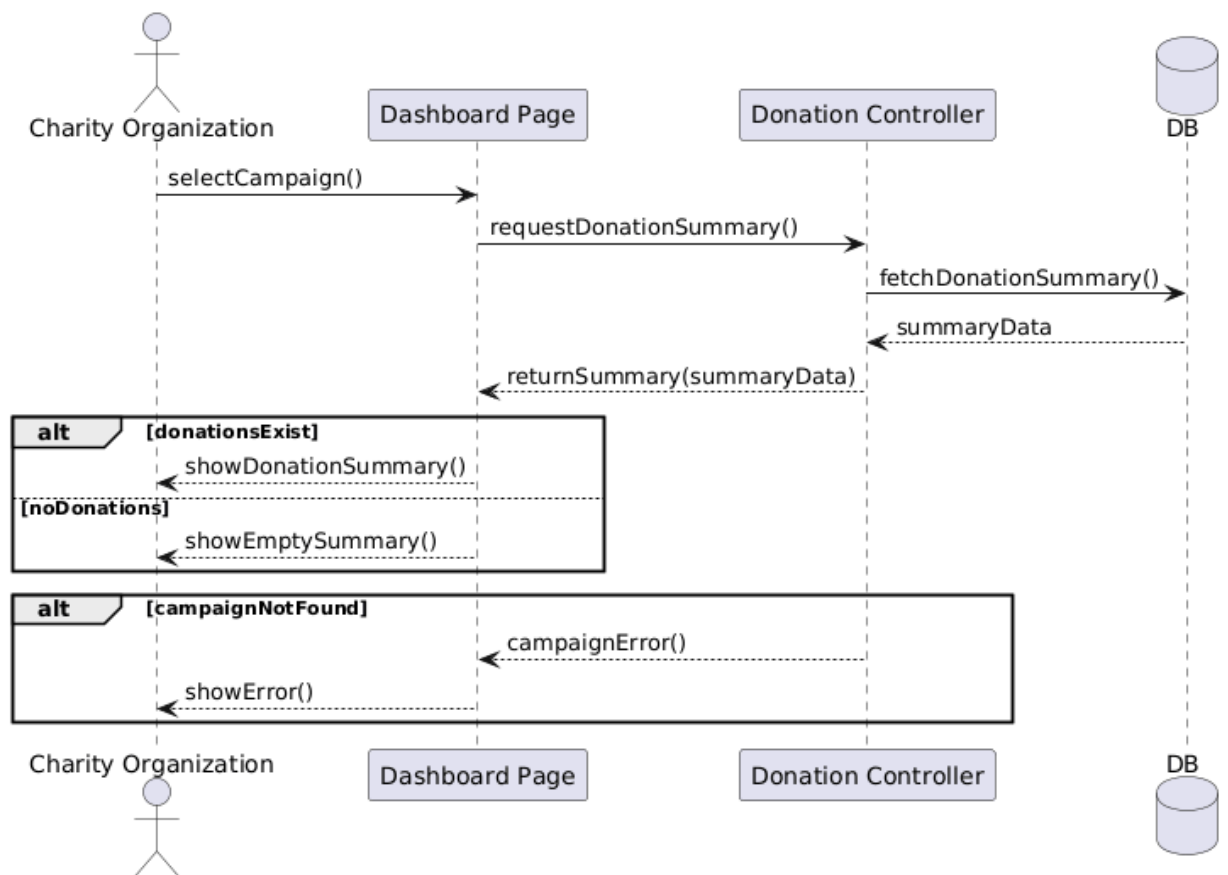


Figure 2.12 View donation summary sequence diagram

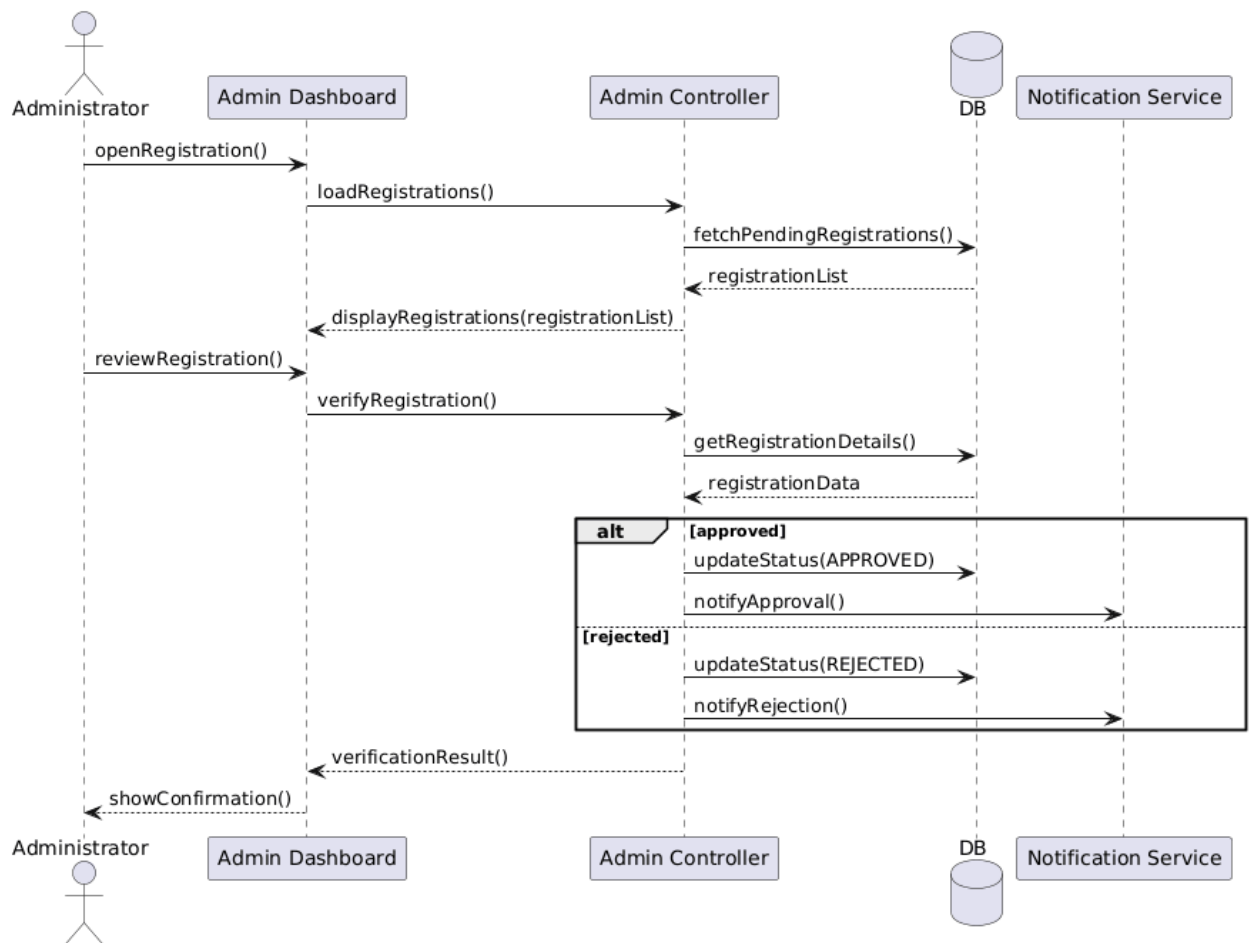


Figure 2.13 Verify Charity registrations sequence diagram

Activity Diagram

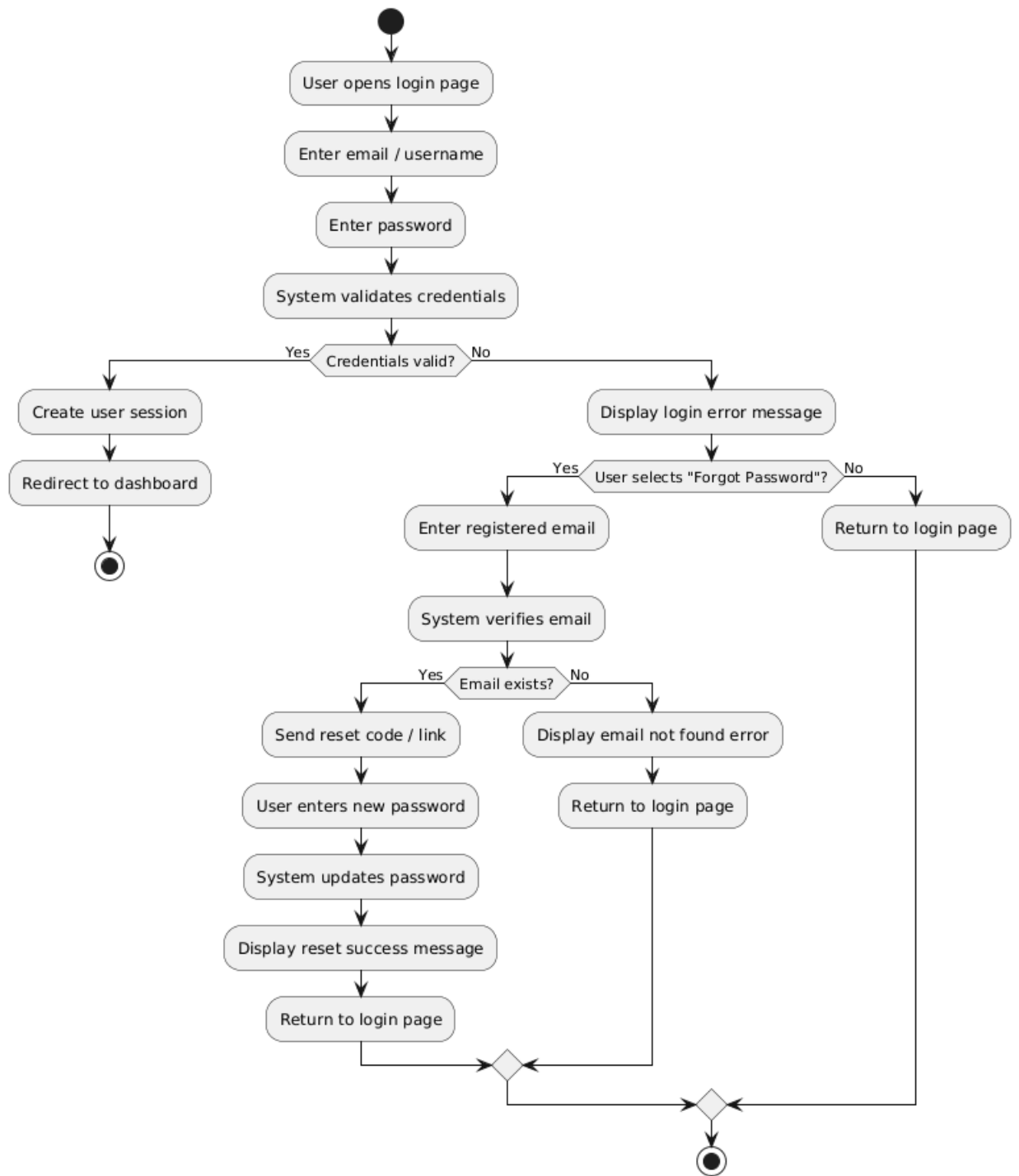


Figure 2.14 Login activity diagram

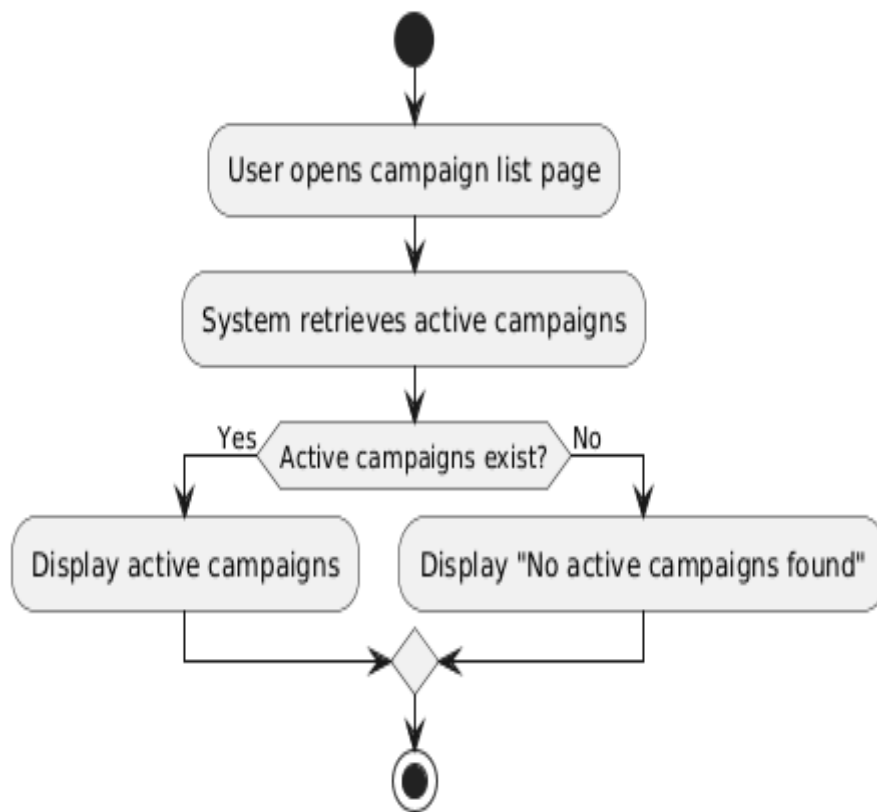


Figure 2.15 View Active Campaigns Activity Diagram



2.16 Search Campaigns Activity Diagram

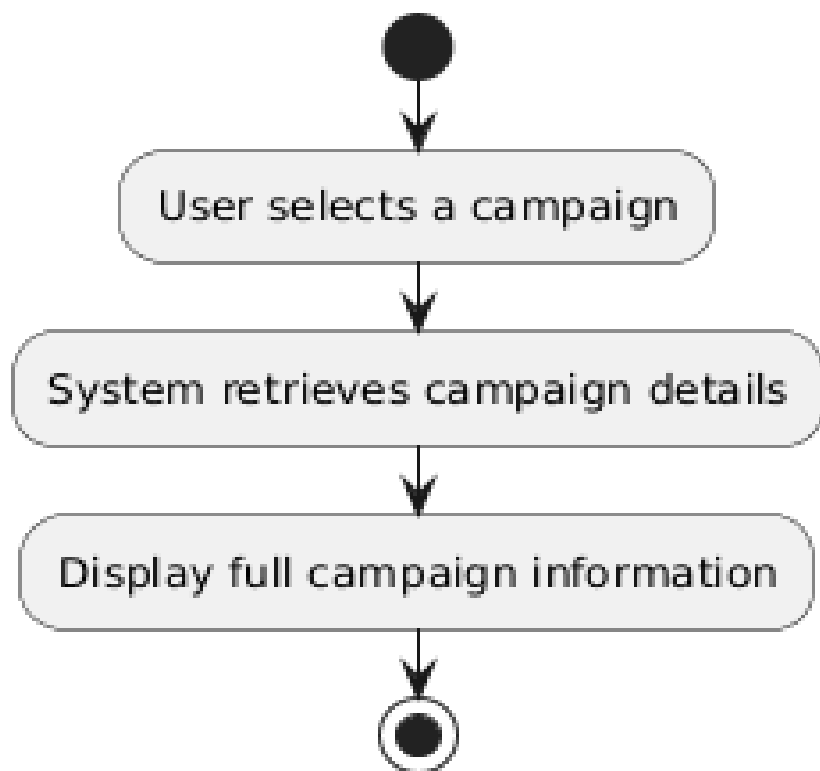


Figure 2.17 View Campaign Details Activity Diagram

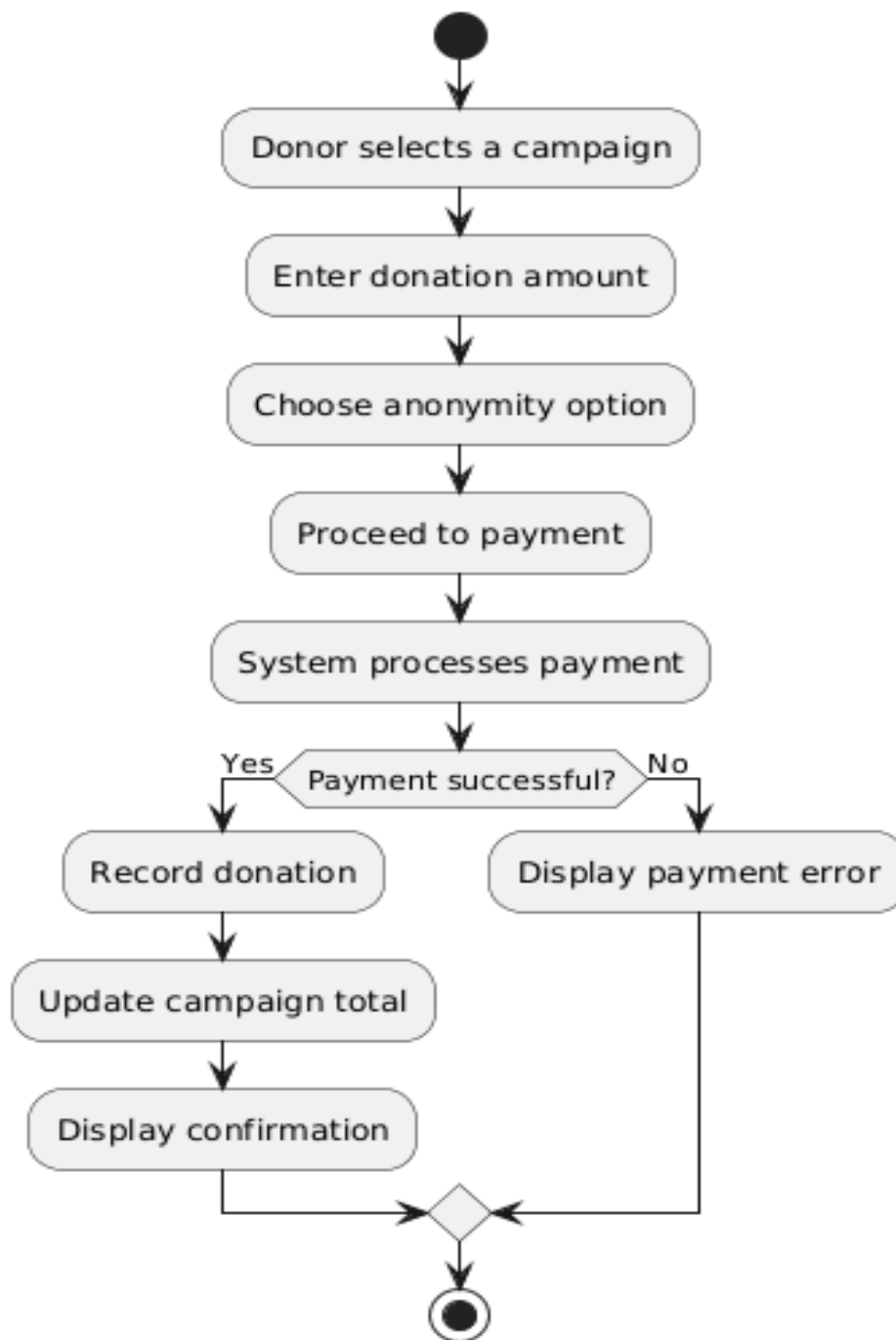


Figure 2.18 Make Donation Activity Diagram

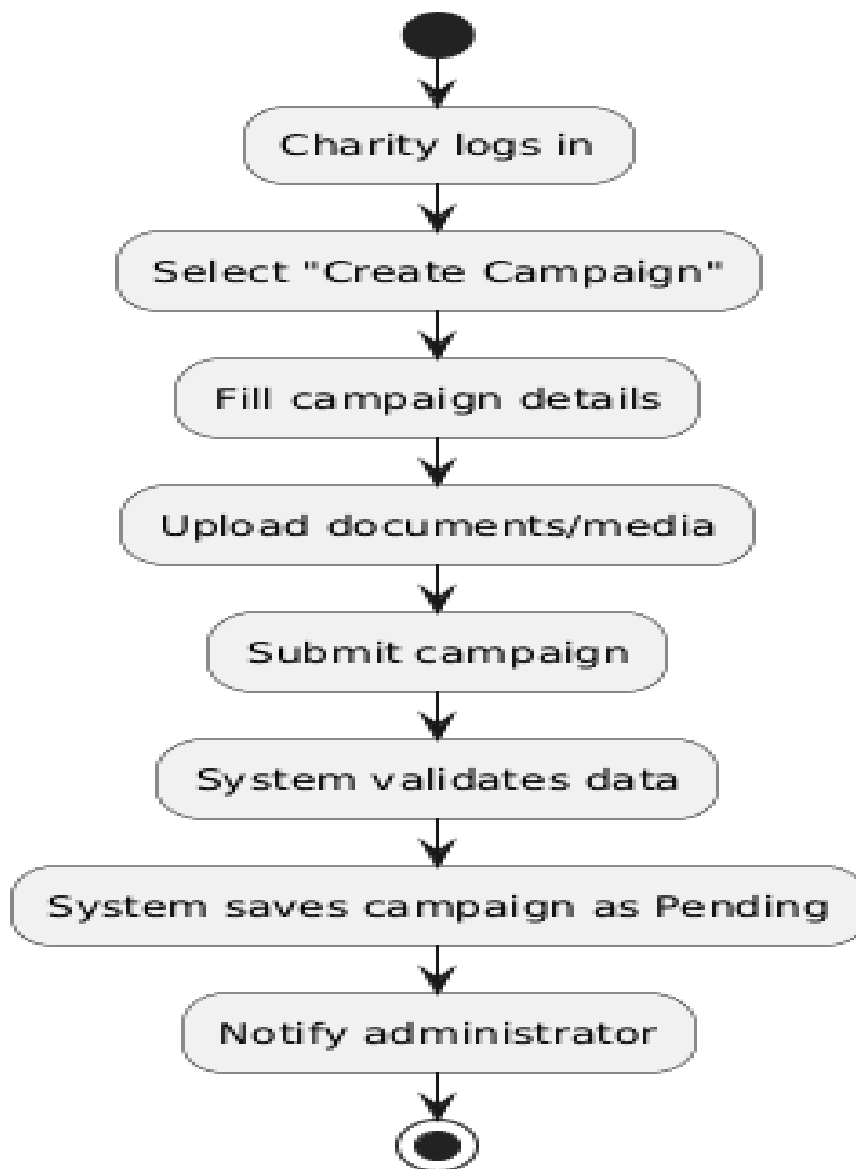


Figure 2.19 Create Campaign Activity Diagram

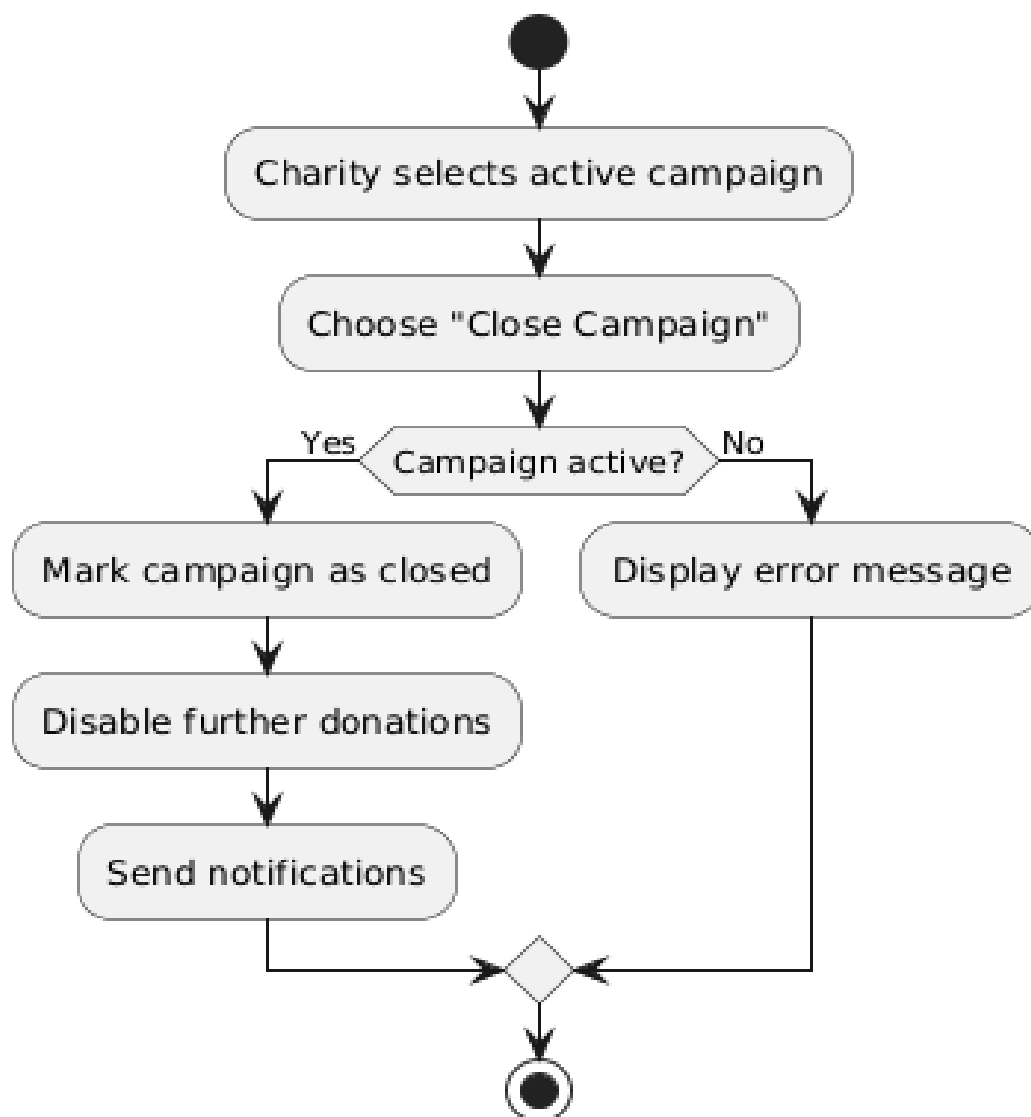


Figure 2.20 Close Campaign Activity Diagram

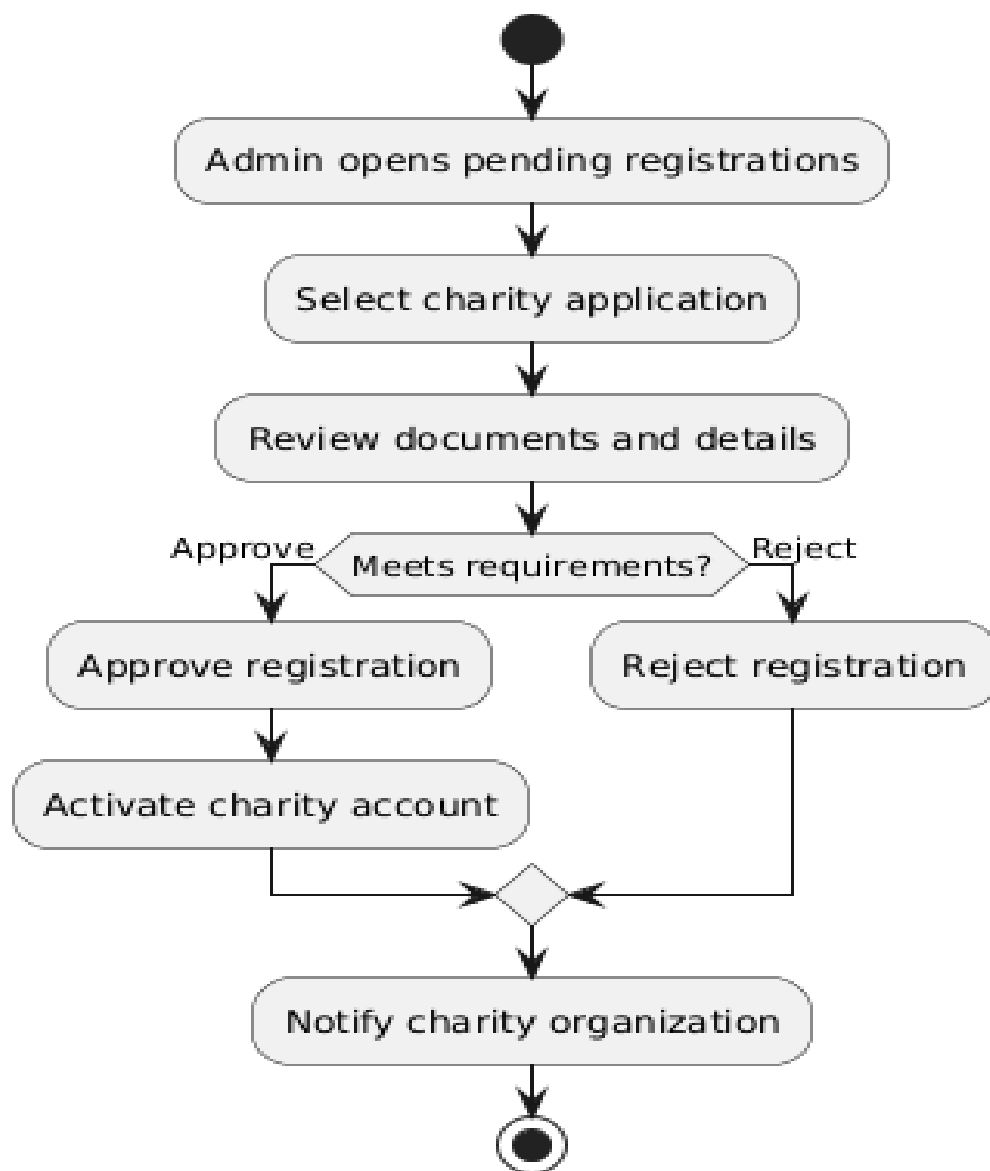


Figure 2.21 Verify Charity Registration Activity Diagram

2.5.3.4 User Interfaces

These are temporary drafts of what our user interface would look.



Figure 2.22 Users landing page UI

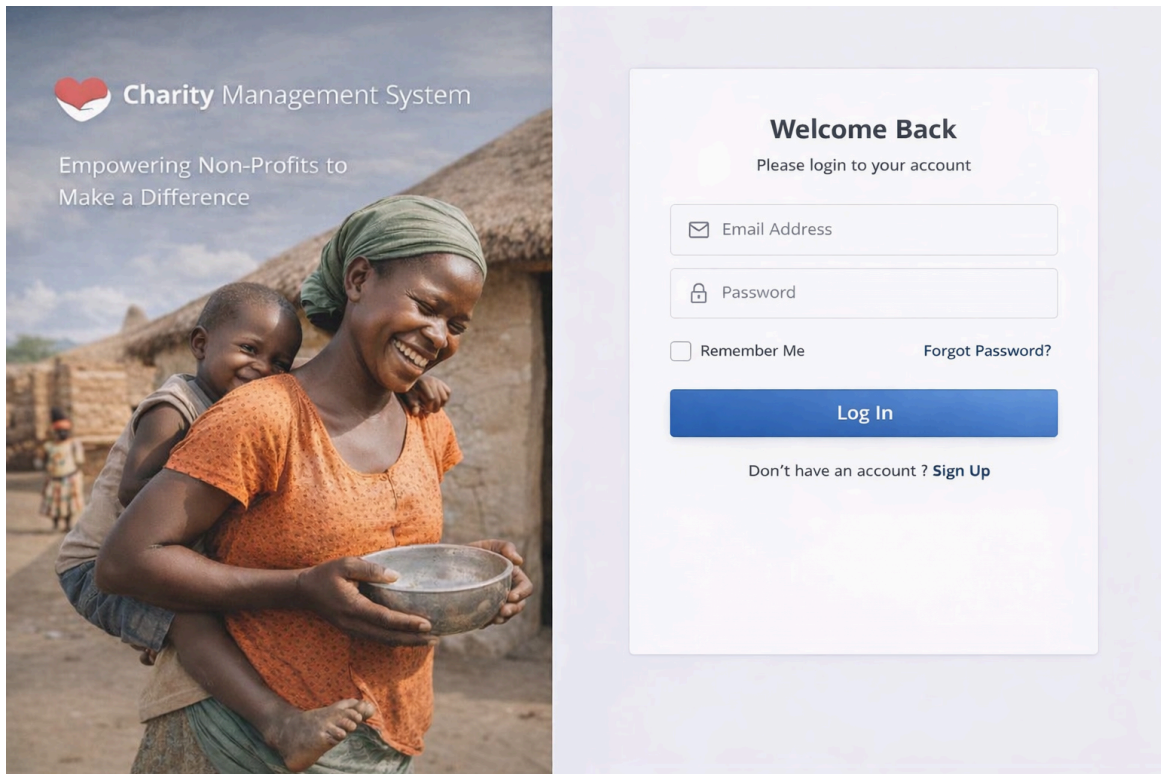


Figure 2.23 Users login page UI

CHAPTER 3

SYSTEM DESIGN

3.1 Introduction

The system design phase provides an overall view of the system from a functional and architectural perspective. It is the process of defining a system's components, modules, interfaces, and data structures in order to satisfy the specified requirements identified during the analysis phase. System design serves as a bridge between requirement analysis and implementation, ensuring that functional and non-functional requirements are translated into a structured and implementable software solution [1].

Effective system design is essential to ensure that all system components operate cohesively to meet user needs and operational constraints. It involves analyzing scalability, performance, reliability, and security requirements while defining clear interfaces and responsibilities for each subsystem. Additionally, the design phase addresses access control mechanisms and data management strategies to protect sensitive information and maintain system integrity. By incorporating these considerations early in the development process, the system becomes more robust, maintainable, and adaptable to future enhancements [2].

3.2. Current Software Architecture

This project proposes a new charity management system and does not replace any existing software. Currently, charity operations are handled manually or through informal tools like spreadsheets and messaging apps, lacking centralization and proper tracking. Therefore, there is no current software architecture to analyze.

3.3 Proposed Software Architecture

3.3.1 Overview

The proposed charity management system adopts a layered software architecture to ensure modularity, maintainability, and clear separation of concerns. This architectural style organizes the system into well-defined layers, where each layer is responsible for a specific aspect of the system's functionality. The layered approach simplifies development and testing while allowing the system to evolve without significantly impacting other components. It is particularly suitable for web-based information systems

that require structured interaction between the user interface, business logic, and persistent data storage .

At a high level, the system is composed of three main layers:

- Presentation Layer
- Application (Business Logic) Layer
- Database Layer

The Presentation Layer provides user interfaces for donors, charity organizations, and administrators, enabling them to interact with the system through web-based views. The Application Layer contains the core business logic, including campaign management, donation processing, notification handling, and access control enforcement. Finally, the Database Layer is responsible for persistent storage of system data such as users, campaigns, donations, and transaction logs.

This architectural design directly supports the functional requirements identified in the requirement analysis phase, such as secure user authentication, campaign creation and management, donation tracking, and administrative oversight. By separating responsibilities across layers, the system enhances scalability, improves security, and reduces coupling between components. Additionally, the layered architecture provides a solid foundation for future enhancements, such as integrating external payment gateways or expanding notification services, without requiring a complete redesign of the system .

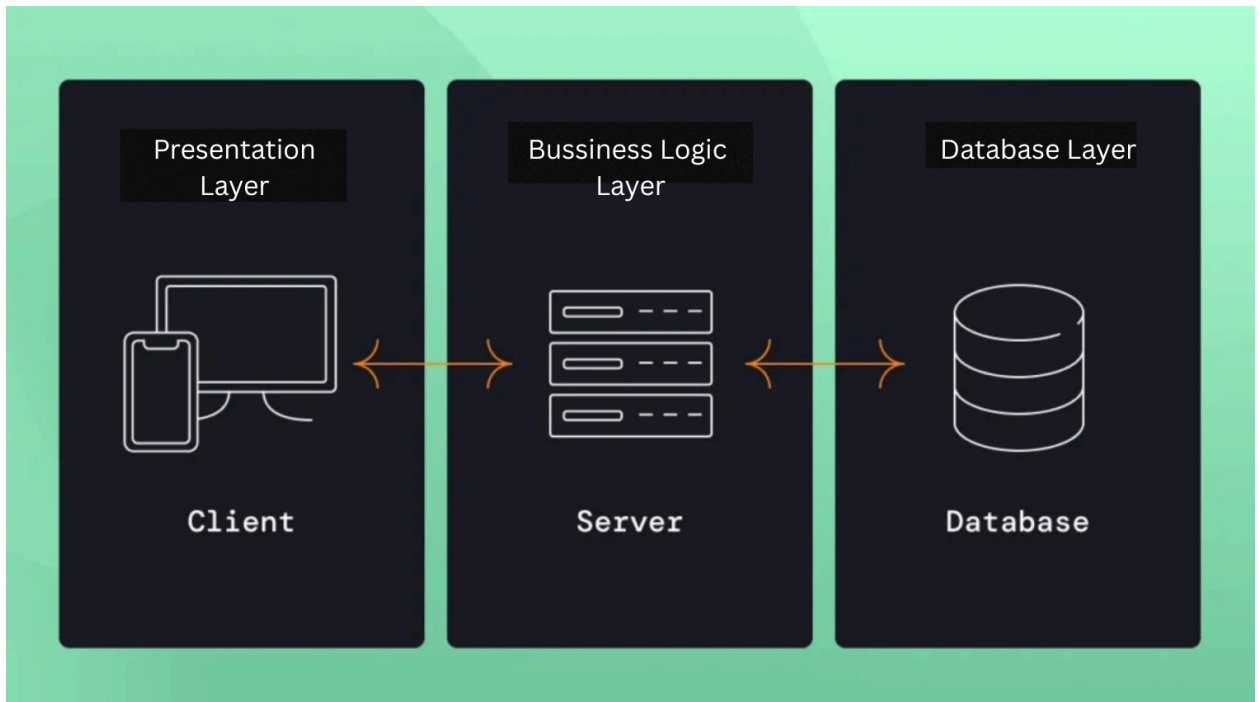


Figure 3.1 Three Tiered Software Architecture

3.3.2 Subsystem Decomposition

Subsystem decomposition is the process of dividing the overall system into smaller, manageable, and logically independent subsystems. Each subsystem is responsible for a specific set of functionalities and encapsulates related operations and data. This approach improves modularity, scalability, maintainability, and clarity of the system design. In the charity management system, the decomposition is based on functional requirements, non-functional constraints, and the roles of system actors.

3.3.2.1 User Management Subsystem

The User Management Subsystem is responsible for handling all user-related operations within the system. This includes user registration, authentication, logout, password reset, and profile management. It supports multiple user roles, namely donors, charity organizations, and administrators, and enforces role-based access control to ensure that users can only access authorized functionalities.

This subsystem plays a critical role in maintaining system security by managing credentials and authentication sessions. It serves as a foundational component upon which other subsystems depend for user identity verification and access control.

3.3.2.2 Campaign Management Subsystem

The Campaign Management Subsystem is responsible for managing the complete lifecycle of charity campaigns within the system. It allows authorized charity organizations to create, edit, and close campaigns while maintaining essential campaign information such as descriptions, funding goals, supporting documents, timelines, and campaign status.

The subsystem also supports administrative workflows by enabling administrators to review, approve, and monitor campaigns to ensure compliance with platform policies. All campaign-related changes are tracked to ensure consistency, transparency, and traceability throughout the campaign lifecycle.

In addition, the Campaign Management Subsystem provides campaign discovery functionality for donors and other users. It enables users to browse, search, and filter available campaigns based on criteria such as keywords, category, location, and campaign status. By integrating discovery features within the same subsystem, the system ensures efficient access to up-to-date campaign information while optimizing read operations for frequent user interactions.

3.3.2.3 Donation Management Subsystem

The Donation Management Subsystem is responsible for recording and managing donation transactions. It allows donors to make contributions, optionally remain anonymous, and view their donation history. The subsystem also supports the generation and retrieval of digital donation receipts.

This subsystem ensures data integrity and accountability by maintaining accurate donation records and transaction logs. It is a critical component for financial transparency and supports both donor reporting and administrative reviewing.

3.3.2.4 Notification Subsystem

The Notification Subsystem manages user subscriptions to campaigns and delivers notifications related to system events. Users can follow campaigns to receive updates on progress, milestones, approvals, and closures.

Notifications may be delivered through email or in-application messages. This subsystem enhances user engagement and ensures timely communication between the system and its users, while remaining loosely coupled with other subsystems through event-based interactions.

3.3.2.5 Administration Subsystem

The Charity Verification and Administration Subsystem supports administrative control and governance of the platform. It enables administrators to verify charity organization registrations, manage user accounts, review donation logs, and transactions.

This subsystem enforces platform rules and ensures compliance with organizational policies. By centralizing administrative functions, it contributes to system integrity, trust, and accountability.

3.3.2.6 Reporting Subsystem

The Reporting Subsystem provides functionality for generating structured reports and summaries related to system activities. It supports the creation of donation summaries, campaign performance reports, and basic statistical overviews based on the data stored in the system.

This subsystem processes data such as donation amounts, campaign progress, and user participation to produce meaningful reports for administrators and authorized stakeholders. Reports may include total donations per campaign, donation trends over a given period, and overall system usage summaries.

By offering accurate and up-to-date reporting capabilities, the Reporting Subsystem enhances transparency and supports informed decision-making without introducing additional system complexity.

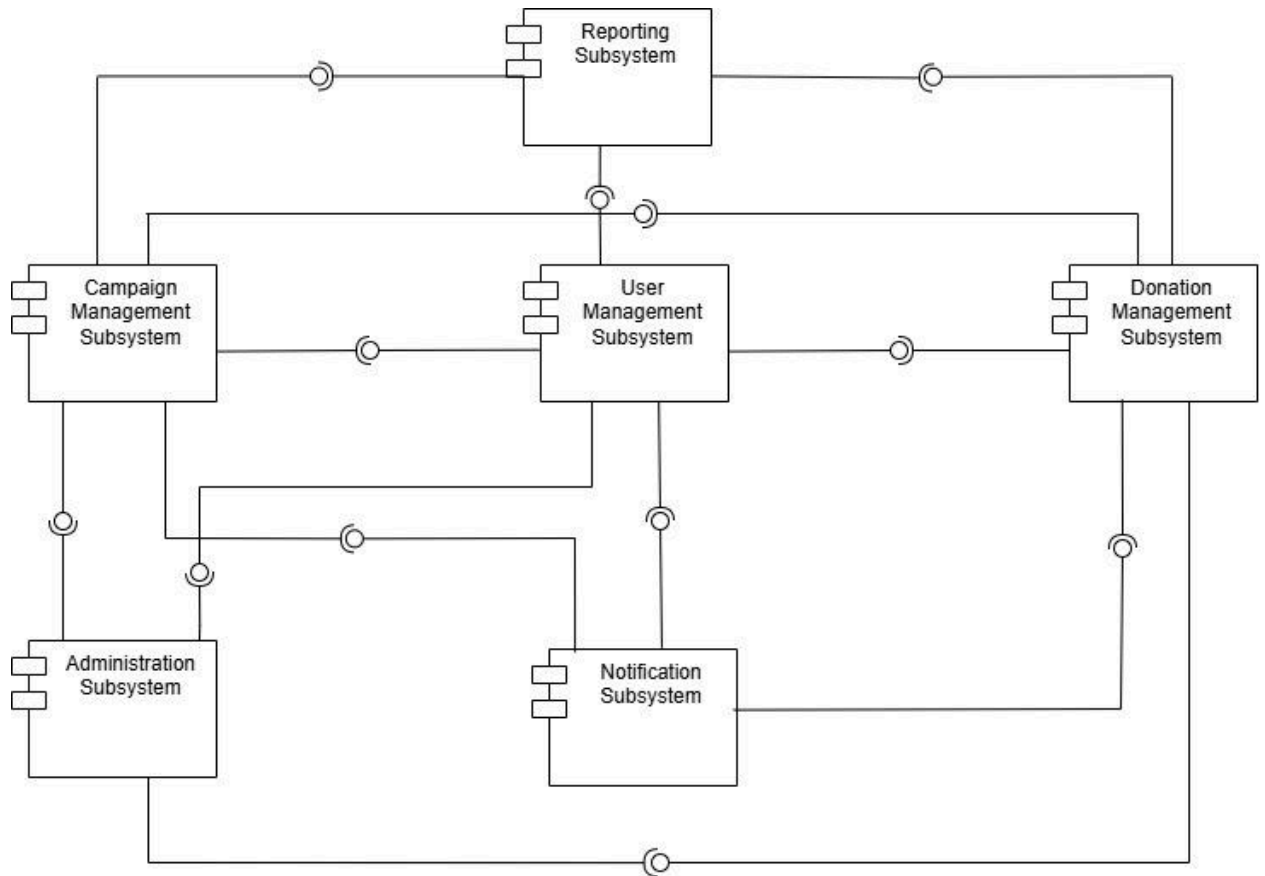


Figure 3.2 Subsystem Decomposition

3.3.3 Hardware/Software Mapping

It describes information such as the system's hardware configuration, the nodes responsible for a particular functionality, and so on. Unified Modeling Language (UML) deployment diagrams are used to depict the Hardware/software mapping. It is used to describe the hardware components where software components are deployed. The Hardware/software mapping is described below.

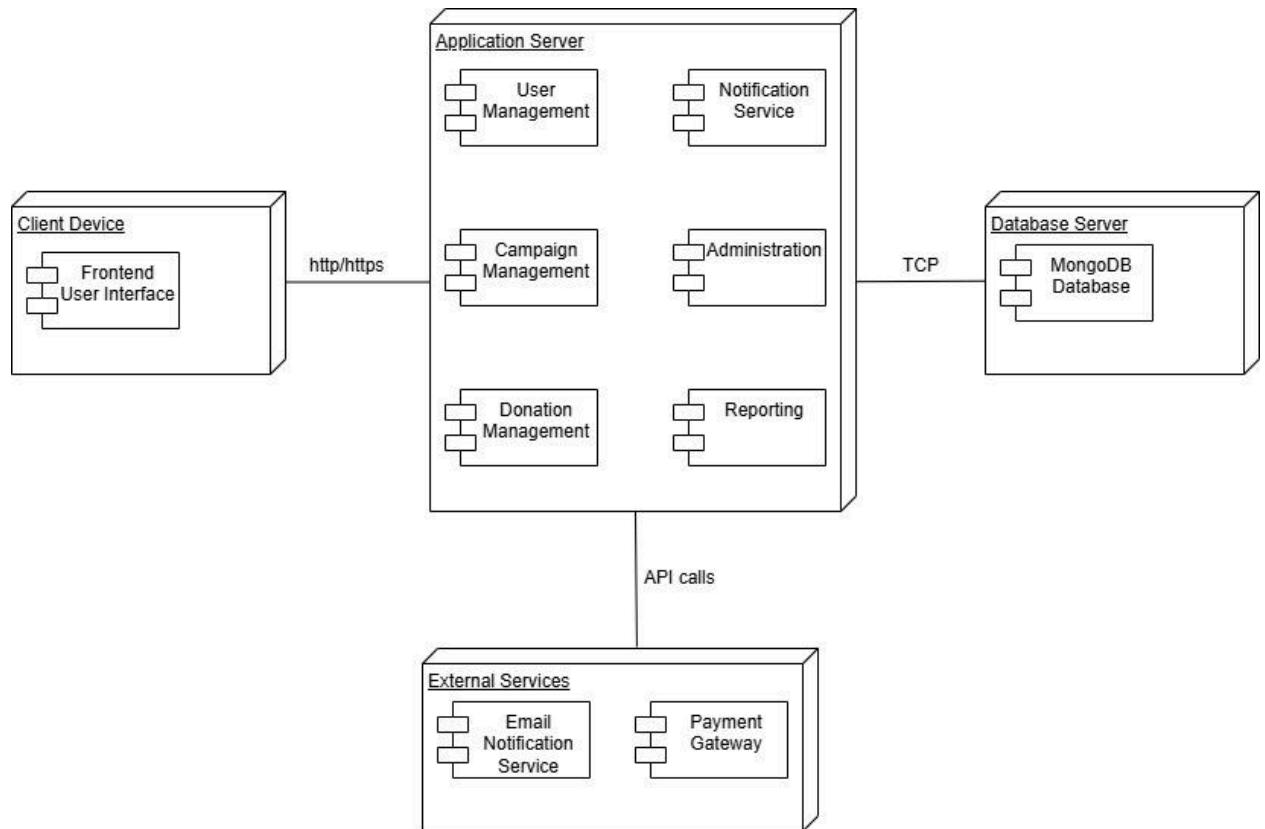


Figure 3.3 Deployment Diagram

3.3.4 Persistent Data Management

The charity management system uses **MongoDB**, a NoSQL document-oriented database, to manage persistent data. MongoDB is selected due to its flexibility, scalability, and suitability for handling semi-structured data. The document-based model allows efficient storage of complex objects such as campaigns, donations, and user profiles without requiring rigid schemas. This flexibility supports frequent updates, evolving requirements, and scalability as the number of users and transactions increases.

Object / Collection	Description
User	Stores donor, charity organization, and administrator account details

Campaign	Stores campaign information created by charity organizations
Donation	Stores donation transactions made by donors
Campaign Document	Stores campaign-related uploaded documents
Notification	Stores system-generated notifications

Table 3.1 Core Data Objects

3.3.5 Access Control and Security

The Charity Management System uses Role-Based Access Control (RBAC) to ensure that system resources are accessed only by authorized users. There are three roles: Donor, Charity Organization, and Administrator, each with permissions tailored to their responsibilities. Donors can browse campaigns, make donations, and view donation history; charities can manage campaigns; administrators can verify charities, review transactions, and configure system settings.

Our system employs JSON Web Tokens (JWT) for user authentication. JWTs are chosen for their compactness, self-contained nature, ease of use, and the ability to be transmitted via HTTP headers. They provide a secure method of authenticating users by including a token with each request. The token's JSON payload can be digitally signed or encrypted, ensuring the secure transmission of information. Additionally, JWTs seamlessly integrate into existing systems and enable user authentication across multiple platforms.

Sensitive operations, including donations, campaign updates, and user management, are logged to ensure accountability. Input validation and access checks protect against unauthorized data manipulation, maintaining the confidentiality, integrity, and availability of system data.

Functionality	Donor	Charity Organization	Administrator
User Registration	✓	✓	✓
User Login / Logout	✓	✓	✓
Reset Password	✓	✓	✓
Manage User profile	✓	✓	✓
View Active Campaigns	✓	✓	✓
View Campaign Details	✓	✓	✓
Search Campaigns	✓	✓	✓
Filter Campaigns	✓	✓	✓
Make Donation	✓	✗	✗
View Donation History	✓	✗	✗
Follow Campaign	✓	✗	✗
Download Donation Receipt	✓	✗	✗
Review Donation Logs	✗	✓	✓
Create Campaign	✗	✓	✗
Edit / Close Campaign	✗	✓	✗
Upload Campaign Documents	✗	✓	✗

Upload Charity Registration Document	X	✓	X
View Donation Summary	X	✓	X
Verify Charity Registration	X	X	✓
Review Financial Transactions	X	X	✓
Configure System Settings	X	X	✓
Receive Notifications	✓	✓	✓

Table 3.2 Access Control

3.4 Detailed Class Diagram

This section offers an in-depth review of the system's classes, along with their dependencies, constraints, attributes, and functions.

1. Donor Side Class Diagram

This class diagram illustrates the core classes involved in the donation process from the donor's perspective. It shows how a donor interacts with charity campaigns by making donations, optionally choosing anonymity. Each donation is associated with a specific campaign and may generate a receipt as confirmation of a successful transaction. The diagram focuses on capturing donation details, campaign progress, and receipt generation while keeping the structure simple and aligned with donor-related system use cases.

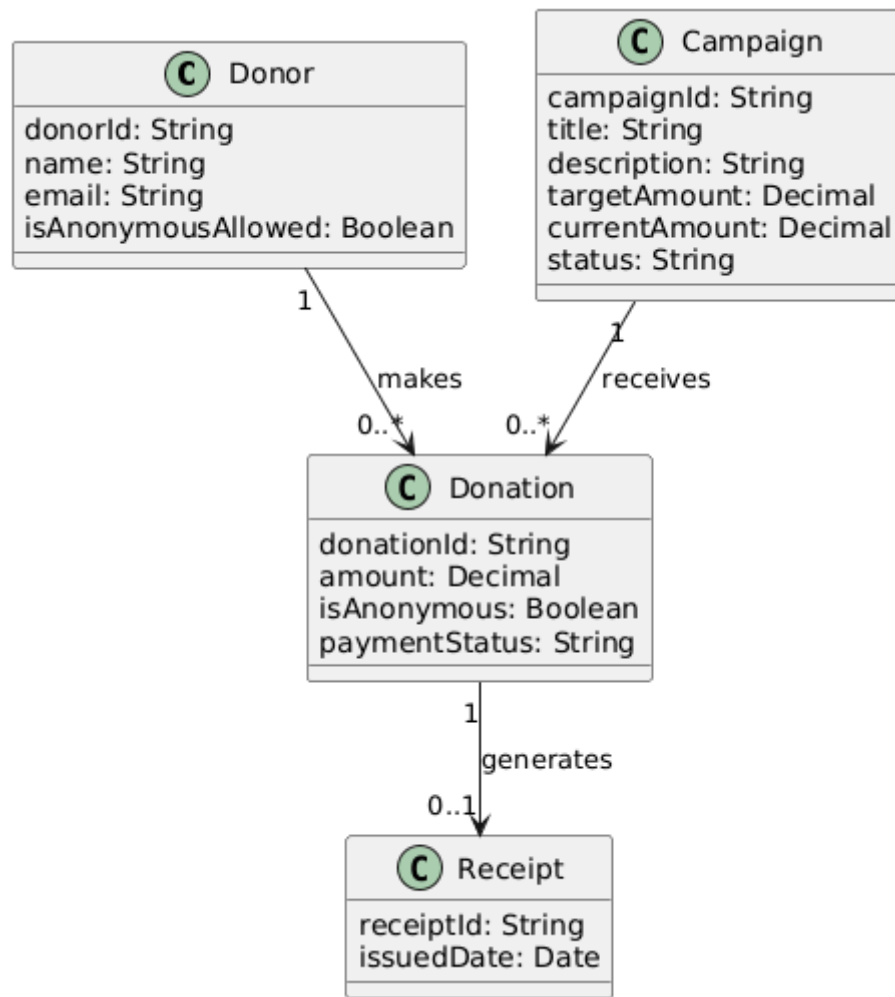


Figure 3.4 Donor Side Class Diagram

2. Charity Organization Side Class Diagram

This class diagram represents the classes used by charity organizations to manage fundraising activities within the system. It highlights how a charity organization creates and manages campaigns and how donations are linked to those campaigns for tracking and reporting purposes. The diagram emphasizes campaign lifecycle management, donation monitoring, and the organization's verification status, reflecting the operational responsibilities of charity organizations in the system.

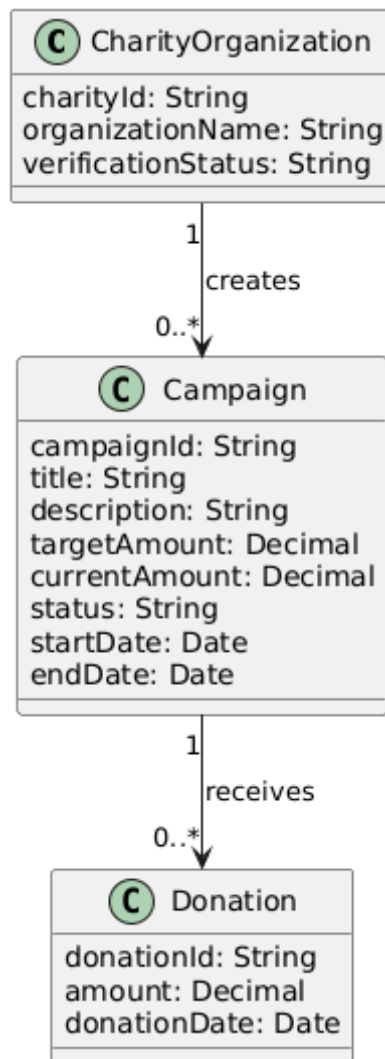


Figure 3.5 Charity Organization Side Class Diagram

3. Administrator Side Class Diagram

This class diagram represents the classes responsible for **administrative management and system oversight**. It illustrates how administrators verify charity organizations, manage user accounts, review system records, and oversee campaigns and donations. The diagram also includes notification-related classes used to communicate important system events to users. Overall, it focuses on **governance, access control, and supervision tasks** performed by administrators to ensure the system operates correctly and transparently.

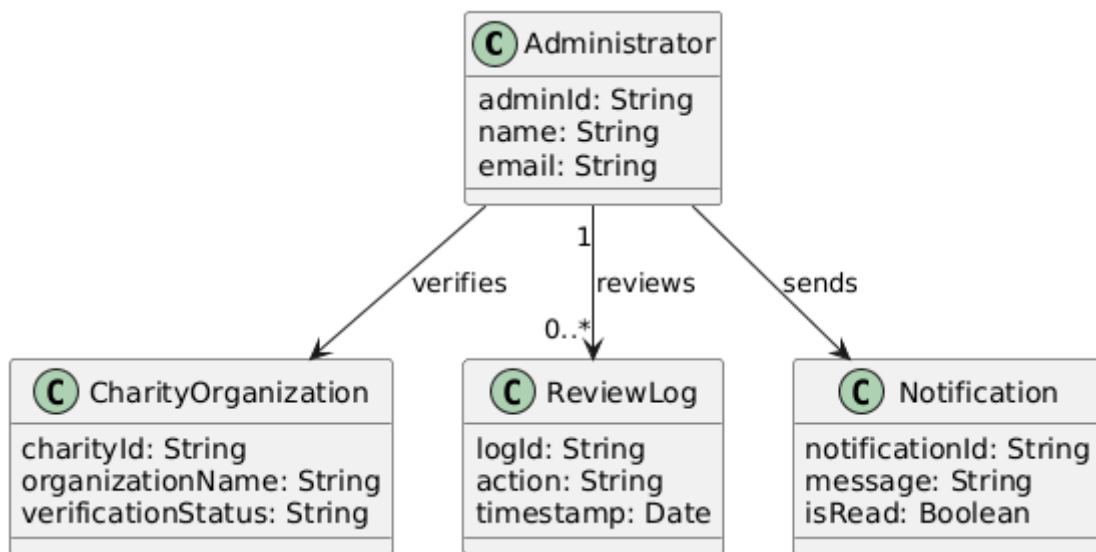


Figure 3.6 Administrator Side Class Diagram

3.5 Packages

1. User Management Package

The User Management package takes care of everything related to users in the system. It allows donors, charity organizations, and administrators to create accounts, log in securely, and manage their profiles. It also controls what each user is allowed to do by assigning roles, making sure everyone only has access to the features meant for them.

2. Campaign Management Package

The Campaign Management package handles all charity campaigns from start to finish. It lets charity organizations create and update campaigns, while administrators review and approve them before they go live. This package makes sure campaign details are accurate, up to date, and clearly visible to donors.

3. Donation Management Package

The Donation Management package manages all donation-related activities in the system. It records each donation, tracks payment status, supports anonymous giving, and generates receipts for donors. This ensures that donations are properly documented and that financial information remains transparent and reliable.

4. Administration Package

The Administration package is used by administrators to oversee and control the platform. It allows them to verify charity organizations, monitor system activities, and

enforce platform rules. This package helps keep the system trustworthy, secure, and well-organized.

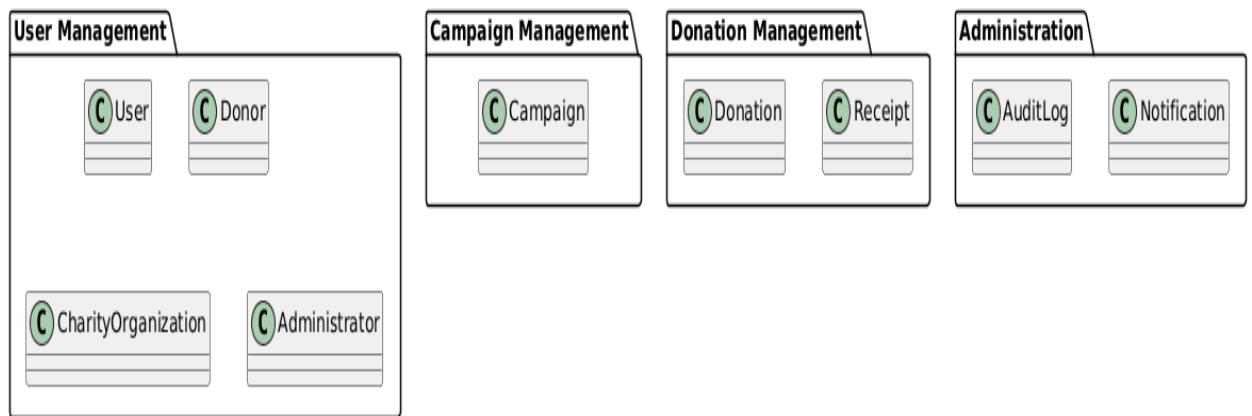


Figure 3.7 Package Diagram

3.5.1 Dependency of Packages

The package dependency structure shows how different parts of the charity management system work together. The User Management package supports both the Campaign Management and Donation Management packages by handling user authentication and role-based access. The Campaign Management package relies on Donation Management to record and track donations made to each campaign. The Administration package connects with all other packages so administrators can verify users, approve campaigns, and review donation activities. This structure keeps responsibilities well separated while allowing the system components to work smoothly together.

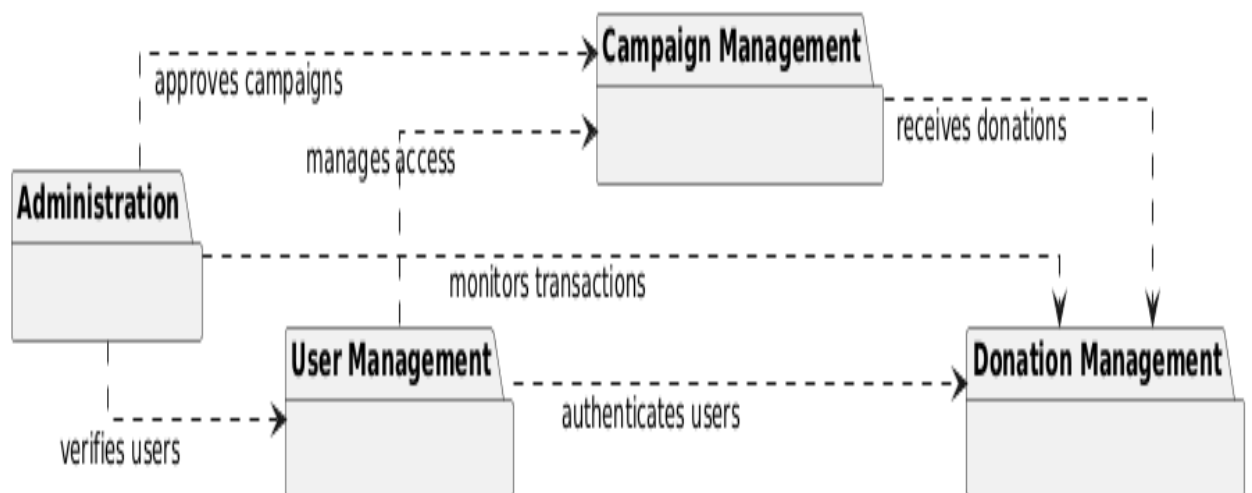


Figure 3.8 Package Dependency Diagram

REFERENCES

- [1] Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2016.
- [2] Roger S. Pressman and Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill Education, 2014.
- [3] Texila International Journal (2025). "Enhancing Non-profit Efficiency and Impact Through Data-Driven Strategies."
- [4] BoardEffect (2024). "Nonprofit embezzlement cases: examples to learn from."
- [5] Unified Modeling Language (UML) Specification
- [6] GoFundMe. *Online Fundraising Platform*
- [7] Charity Navigator. *Charity Accountability and Transparency*.